

NCP-MCI

CERTIFICATION

Multicloud Infrastructure

300 Interview Questions

& Model Answers

AOS 6.x | AHV | Prism | Calm | Flow | Era | NKE

Data Protection | Security | Cloud Platform

Complete Interview Preparation Guide

TABLE OF CONTENTS

Section 1: Nutanix Architecture & Core Concepts	Q 1-10
Section 2: Prism Management & Administration	Q 11-20
Section 3: Storage Management	Q 21-32
Section 4: Networking on Nutanix / AHV	Q 33-42
Section 5: Virtual Machine Management	Q 43-52
Section 6: Data Protection & Disaster Recovery	Q 53-62
Section 7: Scaling, Performance & Optimization	Q 63-72
Section 8: Security & Compliance	Q 73-82
Section 9: Nutanix Calm & Automation	Q 83-92
Section 10: LCM, Upgrades & Troubleshooting	Q 93-102
Section 11: Nutanix Cloud Platform & Xi/NC2	Q 103-112
Section 12: Advanced Topics & Best Practices	Q 113-122
Section 13: Nutanix APIs & Integration	Q 123-132
Section 14: VDI & End User Computing on Nutanix	Q 133-142
Section 15: Exam Preparation & Key Concepts Summary	Q 143-152
Additional Questions (General Topics)	Q 153-300

Section 1: Nutanix Architecture & Core Concepts

Q1 What is Nutanix HCI (Hyper-Converged Infrastructure)?

A Nutanix HCI is a software-defined IT infrastructure that combines compute, storage, and networking into a single integrated system. It eliminates the need for separate SAN/NAS storage arrays by running a distributed storage fabric (DSF) across all nodes in a cluster, providing linear scalability and simplified management through a single pane of glass.

Q2 What is the Nutanix Distributed Storage Fabric (DSF)?

A DSF is Nutanix's core storage layer that pools local drives (SSD + HDD) across all nodes into a single shared storage resource. It provides data redundancy via replication factor (RF), self-healing capabilities, deduplication, compression, and erasure coding — all managed transparently by the Controller VM (CVM).

Q3 What is a CVM (Controller VM)?

A The Controller VM is a dedicated virtual machine running on every Nutanix node that manages all local I/O operations and DSF services. It intercepts and handles all storage I/O, provides data services (dedup, compression, tiering), and communicates with CVMs on other nodes to maintain a unified cluster storage view.

Q4 What hypervisors are supported on Nutanix NCP-MCI?

A Nutanix supports AHV (Acropolis Hypervisor — native, KVM-based), VMware vSphere (ESXi), and Microsoft Hyper-V. AHV is Nutanix's own free hypervisor tightly integrated with AOS for optimal performance and simplified management.

Q5 What is AHV and why is it preferred?

A AHV (Acropolis Hypervisor) is Nutanix's native KVM-based hypervisor included at no extra cost. It is preferred because it offers built-in high availability, live migration (AHV-powered Live Migration), micro-segmentation via Flow, deep integration with Prism, and eliminates the need for additional hypervisor licensing.

Q6 What is the Acropolis Operating System (AOS)?

A AOS is the core software stack that runs on every Nutanix node. It includes the DSF, CVM management, data services (deduplication, compression, erasure coding), cluster management APIs, and the Acropolis layer that manages VMs and storage. AOS abstracts hardware differences and provides a consistent experience across different hypervisors.

Q7 What is a Nutanix node?

A A Nutanix node is an individual server (x86 hardware) that contains CPU, RAM, and local storage (NVMe SSDs, SSDs, HDDs). Each node runs a hypervisor and a CVM. Multiple nodes form a cluster. Nodes are available as all-flash, hybrid, or storage-heavy configurations depending on workload needs.

Q8 What is a Nutanix block?

A A Nutanix block is a physical chassis that holds 1–4 nodes. Blocks are connected via standard 10 GbE or 25 GbE Ethernet switches. Multiple blocks form a cluster. Blocks are available in 1U (1-2 nodes), 2U (2-4 nodes), and other form factors depending on the platform (NX, Dell XC, HPE DX, etc.).

Q9 What is a Nutanix cluster?

A A Nutanix cluster is a group of 3 or more nodes (minimum 3 for RF2) that work together as a single pooled resource. The cluster runs a distributed metadata service, provides unified storage via DSF, and is managed through Prism Element or Prism Central. Clusters can span up to hundreds of nodes for linear scalability.

Q10 What is the minimum number of nodes required for a Nutanix cluster?

A The minimum is 3 nodes for a standard RF2 (Replication Factor 2) cluster. A 2-node cluster is supported with a witness VM for specific use cases like ROBO (Remote Office/Branch Office). RF3 requires a minimum of 5 nodes to tolerate 2 simultaneous node failures.

Section 2: Prism Management & Administration

Q11 What is Prism Element?

A Prism Element (PE) is the local cluster management interface built into each Nutanix cluster. It provides a web-based GUI for managing VMs, storage containers, networks, protection domains, and cluster health. Each cluster has its own Prism Element instance, accessible via the cluster's virtual IP address.

Q12 What is Prism Central (PC)?

A Prism Central is a centralized multi-cluster management platform that allows administrators to manage multiple Nutanix clusters from a single interface. PC provides advanced features like Prism Pro analytics, capacity planning, policy management, Calm orchestration, Flow networking, and X-Play automation.

Q13 What is the difference between Prism Element and Prism Central?

A Prism Element manages a single cluster locally and provides basic operations. Prism Central manages multiple clusters from one interface, adds advanced analytics (Prism Pro), policy-based governance, capacity planning, application orchestration (Calm), and micro-segmentation (Flow). PC is deployed as a VM on one of the managed clusters.

Q14 How do you access Prism Element?

A Access Prism Element by navigating to <https://:9440> in a web browser. You can also access it via any CVM IP on port 9440. Default credentials are admin/admin on first setup, and you are required to change the password on first login for security compliance.

Q15 What is the Nutanix CLI (NCLI)?

A NCLI (Nutanix Command Line Interface) is a command-line tool available on CVMs that provides administrative control over the cluster. You can perform operations like cluster configuration, storage management, VM operations, and health checks. Access NCLI by SSH into any CVM and running the 'ncli' command.

Q16 What is aCLI in Nutanix?

A aCLI (Acropolis CLI) is used to manage AHV hypervisor operations and VM management from the CVM command line. It provides commands for creating/modifying VMs, managing networks, handling snapshots, and live migrations. It is accessed by running 'acli' from the CVM shell.

Q17 What are the key dashboards available in Prism?

A Prism provides: Home (cluster overview, alerts, health score), VM (virtual machine management), Storage (containers, datastores, volume groups), Network (VLANs, managed networks), Hardware (node/disk health), Analysis (performance charts), Alerts (active alerts and events), and Tasks (async operation tracking).

Q18 What is Prism Pro?

A Prism Pro is a licensed add-on for Prism Central that provides advanced analytics, machine-learning-driven capacity forecasting, automated remediation via X-Play, Just-In-Time (JIT) performance insights, custom dashboards, and advanced reporting. It requires an additional license beyond standard Prism Central.

Q19 What is X-Play in Nutanix?

A X-Play (Cross-Play) is Nutanix's no-code automation engine within Prism Pro. It uses playbooks – trigger + action workflows – to automate remediation tasks like adding VM memory when CPU/memory alerts fire, sending notifications, or triggering REST API calls based on performance thresholds and alerts.

Q20 How do you upgrade AOS from Prism?

A Navigate to Prism Element > Settings > Upgrade Software > AOS. Upload the AOS tarball or allow LCM (Life Cycle Manager) to discover and download available versions. The upgrade runs as a rolling upgrade, upgrading one node at a time while keeping VMs running via live migration, ensuring zero downtime.

Section 3: Storage Management

Q21 What is a Storage Container?

A A Storage Container is a logical storage policy unit within the DSF. It maps to a datastore in vSphere or a storage domain in AHV. Each container can have individual settings for replication factor, compression, deduplication, and erasure coding. VMs stored in different containers can have different data efficiency policies.

Q22 What is Replication Factor (RF) in Nutanix?

A Replication Factor (RF) determines how many copies of data are maintained across nodes. RF2 maintains 2 copies, tolerating 1 node failure. RF3 maintains 3 copies, tolerating 2 simultaneous node failures. RF is set per storage container and affects storage utilization (RF2 = ~50% usable, RF3 = ~33% usable).

Q23 What is Erasure Coding (EC-X) in Nutanix?

A EC-X (Erasure Coding) is an alternative to full replication that uses mathematical algorithms to reconstruct data from parity fragments. It provides better storage efficiency than RF2/RF3 (60–80% space savings vs replication) at the cost of slightly higher CPU overhead and rebuild time. Best suited for cold or sequential data.

Q24 What is inline vs post-process deduplication?

A Inline deduplication removes duplicate data blocks as they are written, saving space immediately but adding write latency. Post-process deduplication runs as a background task after data is written, avoiding write latency impact. Nutanix supports both; inline dedup is recommended for VDI workloads with many identical VMs.

Q25 What is compression in Nutanix?

A Nutanix compression reduces data size using LZ4 or Snappy algorithms. Inline compression compresses data before writing to storage (good for all-flash). Post-process compression runs as a background job. Compression is enabled per container and can achieve 30–60% space savings depending on data type.

Q26 What is Storage Tiering in Nutanix?

A Nutanix implements automatic storage tiering (ILM – Intelligent Lifecycle Management) that keeps hot data on SSD (or NVMe) and migrates cold data to HDD automatically. The CVM monitors data access patterns and promotes/demotes data between tiers transparently without manual intervention.

Q27 What is a Volume Group in Nutanix?

A A Volume Group (VG) is a block storage construct that presents iSCSI LUNs directly to VMs or external hosts. VGs bypass the hypervisor storage stack, making them ideal for database workloads (Oracle RAC, SQL Server) that require low-latency block storage and multi-VM shared storage access.

Q28 What is the difference between a container and a volume group?

A A Storage Container provides NFS/SMB-based storage (datastore) to VMs through the hypervisor layer. A Volume Group provides iSCSI block storage directly to VMs, bypassing the hypervisor for lower latency. Containers are used for general VM storage; VGs are used for databases and applications needing raw block access.

Q29 What is Nutanix Files?

A Nutanix Files (formerly ACFS) is a scale-out file server solution that runs as a VM cluster (FSVMs) on Nutanix, providing NFS and SMB file shares. It uses the DSF as its underlying storage and integrates with Active Directory for authentication. It replaces traditional NAS devices with a Nutanix-native file services solution.

Q30 What is Nutanix Objects?

A Nutanix Objects is an S3-compatible object storage service that runs on Nutanix clusters. It stores unstructured data (backups, media, logs) using an object-based model. It provides web-scale scalability, versioning, lifecycle policies, and WORM (Write Once Read Many) compliance for regulatory requirements.

Q31 What is the difference between thick and thin provisioning in Nutanix?

A Thin provisioning allocates storage capacity only as data is written, maximizing storage utilization (default in Nutanix). Thick provisioning pre-allocates the full virtual disk size immediately, guaranteeing space but wasting unused capacity. Nutanix defaults to thin provisioning, and through overcommit, DSF monitors actual capacity used.

Q32 What is the Acropolis Block Services (ABS)?

A ABS provides block storage (iSCSI) natively through Nutanix, allowing external servers or VMs to consume Nutanix storage as iSCSI targets. ABS enables use cases like SQL Server on physical servers, Exchange clusters, and Oracle RAC configurations where shared block storage across multiple servers is required.

Section 4: Networking on Nutanix / AHV

Q33 What is an Open vSwitch (OVS) in AHV?

A Open vSwitch (OVS) is the software-defined virtual switch used by AHV to handle all VM networking. Each AHV host runs OVS, which connects VM vNICs to physical uplinks. OVS supports VLANs, bonding (active-backup, LACP), and integrates with Flow for micro-segmentation and network policy enforcement.

Q34 What is a Nutanix managed network vs. unmanaged network?

A A managed network in AHV has Nutanix managing IPAM (IP address assignment via DHCP) for VMs. An unmanaged network passes VM traffic through to an existing VLAN without DHCP management — VMs get IPs from an external DHCP server. Managed networks simplify IP management for VM deployments and Calm blueprints.

Q35 What is Nutanix Flow?

A Nutanix Flow is a software-defined micro-segmentation and network security solution integrated into AHV. It provides application-centric security policies, allowing administrators to define allow/deny rules at the VM level without requiring additional hardware firewalls. Flow integrates with Prism Central for policy management and Calm for app-aware security.

Q36 What is a security policy in Nutanix Flow?

A A Flow security policy defines the allowed network communications between groups of VMs (categories). Policies operate in monitor mode (logging only) or apply mode (enforcing rules). Three types exist: Application (east-west within an app), Isolation (block traffic between groups), and Quarantine (isolate compromised VMs).

Q37 What is Nutanix Flow Virtual Networking?

A Flow Virtual Networking (formerly Nutanix ANC – Advanced Networking) provides overlay networking capabilities using VXLAN tunnels, enabling multi-tenant isolated virtual networks, virtual routers, floating IPs, VPN connectivity, and load balancing without requiring changes to the physical network infrastructure.

Q38 What is a bond in AHV networking?

A A bond (NIC teaming) in AHV combines multiple physical NICs for redundancy and throughput. AHV supports active-backup (failover only), balance-slb (load balancing without LACP), and balance-tcp (requires LACP/802.3ad on the switch). The default is active-backup with two NICs, providing network redundancy for every host.

Q39 What are Nutanix Categories used for in networking?

A Categories are key-value metadata tags applied to VMs/applications in Prism Central. In Flow, categories define policy groups – VMs in the same category share security policies. Example: Category 'AppType=WebServer' groups all web tier VMs; Flow policies then control which categories can communicate with each other.

Q40 What port does Prism use for HTTPS?

A Prism uses port 9440 for HTTPS web access. Additionally, port 2222 is used for CVM SSH, port 2100 is the Prism gateway, and port 9440 is used for inter-CVM communication as well. For Prism Central to Prism Element communication, port 9440 is also the primary port.

Q41 How does AHV handle VM live migration?

A AHV performs live migration using a memory pre-copy technique: the running VM's memory is copied iteratively to the target node while the VM continues running. During final cutover, the VM is briefly paused (<1 second), remaining memory pages are copied, and the VM resumes on the destination node. Storage stays in DSF (no storage migration needed).

Q42 What is the purpose of the backplane network in Nutanix?

A The backplane (or storage) network handles CVM-to-CVM communication, DSF replication traffic, and storage I/O between nodes. It should be on a dedicated VLAN or bond to separate storage traffic from VM/management traffic. High-throughput, low-latency NICs (10/25 GbE) are recommended for the backplane network.

Section 5: Virtual Machine Management

Q43 How do you create a VM in Prism Element (AHV)?

A Go to Prism Element > VM > +Create VM. Enter VM name, vCPU count, memory (GiB), and disk configuration. Add a disk (clone from image or create new vDisk), configure network (select managed/unmanaged network). Optionally attach a CD-ROM (ISO image). Click Save. Power on the VM and access it via the console.

Q44 What is a Nutanix Disk Image?

A A Disk Image is a pre-built virtual disk (QCOW2, VMDK, VHD, ISO) stored in the Image Service. Images serve as templates for VM disk cloning. The Image Service in Prism Central manages a centralized image repository, replicating images to multiple clusters for fast VM deployment across the environment.

Q45 What is the Image Service in Nutanix?

A The Image Service is a centralized catalog of OS images, virtual disks, and ISO files used for VM provisioning. Images are stored in the DSF and can be replicated across clusters. In Prism Central, the Image Service is global — an image uploaded once is available to all registered clusters (with optional placement policies).

Q46 How do snapshots work in Nutanix AHV?

A Nutanix snapshots use a redirect-on-write mechanism at the storage layer (crash-consistent by default). When a snapshot is taken, subsequent writes go to new storage extents while the snapshot preserves the original data pointers. Snapshots are near-instantaneous, regardless of VM size, since they only record metadata changes.

Q47 What is the difference between a snapshot and a clone in Nutanix?

A A snapshot preserves the VM state at a point in time for recovery purposes — it is not a standalone VM. A clone creates a fully independent VM copy using linked clones (space-efficient, sharing base disk blocks) or full clones. Nutanix uses space-efficient clones by default, enabling rapid deployment of many VMs from a single image.

Q48 How do you perform a VM live migration in AHV?

A In Prism Element, select the VM > Actions > Migrate. Choose the target host. The live migration copies memory pages to the target host while the VM runs, pauses briefly for cutover, and resumes on the new host. Alternatively, use aCLI: 'acli vm.migrate host='. No storage migration is required since data lives in DSF.

Q49 What is VM High Availability (HA) in Nutanix?

A Nutanix HA automatically restarts VMs on surviving nodes when a node fails. For AHV, HA is enabled by default and requires sufficient free capacity (best practice: N+1 or N+2 headroom). The cluster detects a node failure within seconds and begins restarting VMs on healthy nodes using the pre-configured HA reservation.

Q50 What is VM affinity/anti-affinity in Nutanix?

A VM-to-host affinity rules pin VMs to specific hosts or host groups (useful for licensing, NUMA, or compliance). VM-to-VM anti-affinity rules ensure VMs don't run on the same host (for HA distribution). In Prism Central, these are configured under the VM's settings or via category-based placement policies.

Q51 What is the Guest Tools (NGT) in Nutanix?

A Nutanix Guest Tools (NGT) is a software package installed inside VMs that enables advanced capabilities: application-consistent snapshots (VSS on Windows), self-service file restore via Nutanix Files, VM mobility scripts, and communication between the CVM and the guest OS for enhanced monitoring and management.

Q52 What is vDisk and how does it differ from a physical disk?

A A vDisk is a virtual disk image stored in the DSF as a series of data extents distributed across nodes. Unlike physical disks, vDisks are software-defined, support thin provisioning, can be dynamically resized online, and benefit from DSF services (dedup, compression, replication). Multiple vDisks can be attached to a single VM.

Section 6: Data Protection & Disaster Recovery

Q53 What is a Protection Domain in Nutanix?

A A Protection Domain (PD) is a group of VMs/storage entities protected together by consistent snapshot and replication schedules. PDs define RPO (Recovery Point Objective) via snapshot frequency and RTO via replication to remote sites. They form the basis of Nutanix's built-in async disaster recovery capabilities.

Q54 What is Nutanix Leap?

A Nutanix Leap is a disaster recovery solution that provides policy-driven DR orchestration for AHV workloads. It supports one-click failover, automated runbooks, and integration with Xi Cloud Services for cloud-based DR. Leap manages Recovery Plans with test failover capabilities and dependency sequencing for multi-tier applications.

Q55 What is Recovery Plan in Nutanix Leap?

A A Recovery Plan in Leap defines the ordered failover/failback sequence for protected VMs, including boot order, IP address mapping (test vs. production network), and stage delays. Plans support test failover (non-disruptive), planned failover (graceful), and unplanned failover (disaster recovery) scenarios.

Q56 What is the difference between async, NearSync, and sync replication?

A Async replication: RPO 1 hour minimum, data replicated periodically, lower bandwidth. NearSync: RPO 20 seconds to 1 minute using lightweight snapshots, better RPO with moderate bandwidth. Synchronous replication: RPO 0 (zero data loss), every write confirmed at both sites before acknowledging to guest, requires low-latency (<5ms RTT) network.

Q57 What is Nutanix Xi Leap?

A Xi Leap is Nutanix's cloud-based DR service hosted in Nutanix-operated cloud datacenters (Xi Cloud Services). It provides DR-as-a-Service for on-premises Nutanix clusters, allowing failover to Xi Cloud without owning a secondary datacenter. Xi Leap handles replication management and provides compute resources for failed-over workloads.

Q58 What are the key RPO and RTO values supported by Nutanix DR?

A Nutanix supports: Async DR (RPO 1 hour, RTO 10–30 minutes), NearSync (RPO 20 seconds, RTO minutes), Synchronous (RPO 0, RTO seconds for Metro Availability). Actual RTO depends on workload count, storage size, and network bandwidth. Leap Recovery Plans automate failover to improve RTO consistency.

Q59 What is Nutanix Metro Availability?

A Metro Availability provides synchronous replication between two Nutanix clusters across metro distances (up to 5ms RTT). It creates an active-active stretched cluster where VMs run at one site, with every write synchronously replicated to the secondary site. If the primary site fails, VMs automatically restart at the secondary site with zero data loss.

Q60 What is a Remote Site in Nutanix data protection?

A A Remote Site is a configured replication target — another Nutanix cluster or Nutanix cloud endpoint — to which Protection Domain snapshots are replicated. Remote Sites are configured in Prism Element (Data Protection > Remote Sites) with network settings, bandwidth throttling, and compression options for WAN-efficient replication.

Q61 What is the difference between crash-consistent and app-consistent snapshots?

A Crash-consistent snapshots capture the raw disk state — equivalent to a sudden power loss — suitable for most VMs. Application-consistent snapshots (enabled via NGT + VSS on Windows) flush in-flight writes and quiesce the application before snapping, ensuring a clean state for databases and exchange servers for reliable recovery.

Q62 What is Nutanix Mine?

A Nutanix Mine is an integrated backup solution that combines Nutanix HCI infrastructure with partner backup software (Veeam, HYCU, Commvault) in a pre-validated configuration. Mine provides a dedicated backup target running on Nutanix nodes, consolidating primary and backup infrastructure to reduce hardware sprawl and simplify backup management.

Section 7: Scaling, Performance & Optimization

Q63 How do you scale a Nutanix cluster?

A Scale a Nutanix cluster by adding nodes (scale-out) or adding larger nodes (scale-up). To add a node: physically install it in the rack, connect network cables, and use the Expand Cluster wizard in Prism Element. The cluster automatically redistributes data across new nodes. No downtime is required during node expansion.

Q64 What is the N+1 and N+2 HA reservation?

A N+1 reserves capacity equivalent to the largest node's resources, ensuring that if one node fails, all its VMs can be restarted on remaining nodes. N+2 reserves enough capacity for two node failures simultaneously. Best practice is to configure N+1 at minimum and N+2 for critical workloads to maintain HA during failures.

Q65 What is the purpose of the SSD tier in Nutanix?

A The SSD tier (or NVMe tier) serves as both a read cache (absorbing hot read data) and the primary write path for all new data. All writes are first committed to SSD for low latency, then drained to HDD (in hybrid nodes) during low-I/O periods. The SSD tier provides consistent sub-millisecond latency for active workloads.

Q66 What is All-Flash vs. Hybrid configuration?

A All-Flash nodes contain only NVMe/SSDs, providing maximum IOPS, lowest latency, and highest throughput for performance-intensive workloads (databases, VDI). Hybrid nodes contain both SSDs (cache/active tier) and HDDs (capacity tier), offering more cost-effective storage for mixed or capacity-intensive workloads with moderate performance.

Q67 What is the role of the CVM in I/O handling?

A All VM I/O is intercepted by the CVM on the local node via a storage controller emulation (SCSI pass-through). The CVM processes the I/O using DSF — it checks the local SSD first (data locality), serves cache hits immediately, writes new data to local SSD, and replicates to remote CVMs for RF consistency — all before acknowledging to the VM.

Q68 What is data locality in Nutanix?

A Data locality means that a VM's data is preferentially stored on the same node where the VM runs. This minimizes network I/O by serving most reads from the local CVM's SSD/HDD. When a VM migrates to another node, the CVM gradually migrates the VM's data to the new node's local storage in the background.

Q69 How does Nutanix handle disk failure?

A When a disk fails, the CVM detects the failure and marks the disk as bad. DSF automatically rebuilds the missing data replicas using surviving copies across other nodes (self-healing). The rebuild uses spare capacity across the cluster. An alert fires in Prism, and the failed disk should be replaced following hot-swap procedures.

Q70 What is the impact of adding an all-flash node to a hybrid cluster?

A Adding all-flash nodes to a hybrid cluster creates a mixed cluster. Best practice is to keep clusters homogeneous (all-flash or all-hybrid) for predictable performance. In mixed configurations, storage policies and tiering behavior may be unpredictable. Nutanix recommends separate all-flash and hybrid clusters for distinct workloads.

Q71 What is the Nutanix Performance Monitoring approach in Prism?

A Prism Analysis provides real-time and historical performance charts for VMs, storage containers, and hosts. Key metrics include IOPS, latency (read/write), throughput, CPU%, memory%, and storage I/O controller time. Prism Pro adds anomaly detection, predictive performance alerts, and What-If capacity planning scenarios.

Q72 What is the NCC Health Check?

A NCC (Nutanix Cluster Check) is a diagnostic framework that runs hundreds of health checks against the cluster: hardware health, software versions, configuration best practices, disk/memory/network status, and security compliance. NCC runs automatically on a schedule and can be triggered manually from Prism or via NCLI: 'ncli health check run-all'.

Section 8: Security & Compliance

Q73 What is RBAC in Nutanix?

A Role-Based Access Control (RBAC) in Prism Central restricts user access based on assigned roles. Built-in roles include Cluster Admin, Prism Admin, User Admin, Viewer, and Developer. Custom roles can be created with granular permissions. RBAC integrates with Active Directory/LDAP for enterprise SSO and centralized identity management.

Q74 What are the default roles in Prism Central?

A Default roles: Super Admin (full access), Prism Admin (cluster management), Cluster Admin (single cluster), User Admin (user management), Developer (Calm automation), and Viewer (read-only). Roles are assigned to users or AD groups and can be scoped to specific clusters or categories for least-privilege access.

Q75 What is Security Configuration Manager (SCMA) in Nutanix?

A SCMA is Nutanix's automated security hardening tool that enforces CIS/STIG compliance standards on AHV hosts and CVMs. It continuously monitors security configurations and automatically remediates deviations. SCMA helps organizations maintain compliance with DoD STIGs, PCI-DSS, and HIPAA security baselines without manual intervention.

Q76 What is Data-at-Rest Encryption (DARE) in Nutanix?

A DARE encrypts all data stored on Nutanix nodes using AES-256 encryption. Two approaches: Software Encryption (SEDs not required, managed by Nutanix key management) and Self-Encrypting Drives (SEDs) with external Key Management Server (KMS) integration. DARE ensures data is unreadable if drives are physically removed from nodes.

Q77 What is the Nutanix Key Management Server (KMS)?

A Nutanix integrates with external KMIP-compliant KMS solutions (Thales, Vormetric, IBM SKLM) for managing encryption keys used by DARE. The KMS stores and manages encryption keys separately from the encrypted data, providing separation of duties and enabling key rotation without data re-encryption. Nutanix's own Local KMS is also available.

Q78 What are Nutanix security categories?

A Security categories in Nutanix context refer to Flow security policy categories — metadata tags applied to VMs to group them for micro-segmentation policies. For governance, categories like Environment (Prod/Dev), AppType, and Tier (Web/App/DB) are used to apply consistent security policies across VM groups without IP-based rules.

Q79 What is the CVM security hardening approach?

A CVMs run a minimal, hardened Centos-based OS with only necessary services enabled. Nutanix regularly publishes security advisories and AOS patches. Best practices include: restricting SSH access, enabling 2FA, using RBAC, enabling audit logging, configuring NTP, using SSL certificates (replacing default self-signed certs), and running NCC security checks.

Q80 What is Two-Factor Authentication (2FA) in Nutanix?

A Nutanix Prism supports 2FA via TOTP (Time-based One-Time Password) using authentication apps or integration with RADIUS servers for hardware token support. 2FA is configured in Prism Settings > Authentication. When enabled, users must provide their password plus a one-time code to access Prism Element or Prism Central.

Q81 What is audit logging in Nutanix?

A Nutanix logs all administrative actions (VM operations, configuration changes, user logins) in an audit log accessible from Prism (Settings > Audit Logs) or via REST API. Logs can be forwarded to SIEM solutions via syslog. Audit logs record the user, action, timestamp, and result for compliance and forensic investigation purposes.

Q82 What is the Nutanix Security Development Lifecycle?

A Nutanix follows a Security Development Lifecycle (SDL) incorporating threat modeling, secure code reviews, static analysis (SAST), dynamic testing (DAST), penetration testing, and vulnerability disclosure programs. Security patches are released via AOS/AHV updates, and critical CVEs are addressed with emergency patches outside regular release cycles.

Section 9: Nutanix Calm & Automation

Q83 What is Nutanix Calm?

A Nutanix Calm is an application lifecycle management and orchestration platform integrated with Prism Central. It provides blueprint-based application deployment across multi-cloud and on-premises environments, runbook automation, policy governance, and self-service portals. Calm supports AHV, VMware, AWS, Azure, GCP, and bare-metal deployments.

Q84 What is a Calm Blueprint?

A A Calm Blueprint is a declarative template that describes an application's infrastructure (VMs, networks, storage), configuration (OS, software, scripts), and lifecycle actions (install, update, scale-in/scale-out, delete). Blueprints are version-controlled, shareable, and can deploy consistent application stacks across multiple cloud targets.

Q85 What types of blueprints exist in Calm?

A Calm supports: Single VM Blueprints (simple one-VM applications), Multi-VM Blueprints (complex multi-tier apps with dependencies), and MPI Blueprints (for distributed HPC jobs). Blueprints can include cloud-init scripts, shell/PowerShell scripts, Ansible playbooks, and Chef/Puppet integration for configuration management.

Q86 What is a Calm Action?

A Actions in Calm define operations that can be performed on an application or individual service. System actions (Create, Delete, Start, Stop, Restart) are built-in. Custom actions (Upgrade, Backup, Scale) can be user-defined with scripted tasks. Actions can be triggered manually, via runbooks, or on schedules.

Q87 What is a Calm Marketplace?

A The Calm Marketplace is a self-service catalog where published blueprints are available for end users to launch without needing to understand the underlying infrastructure. Administrators publish blueprints to the marketplace with cost estimates, descriptions, and access controls. Users can request app deployments via a shopping-cart-style interface.

Q88 What is a Runbook in Nutanix Calm?

A A Runbook is a reusable automation script in Calm that performs infrastructure operations without deploying a full application. Examples: adding/removing users, running maintenance scripts, updating configurations across multiple VMs, or responding to monitoring alerts. Runbooks support endpoints and can target VMs not managed by Calm blueprints.

Q89 How does Calm handle multi-cloud deployments?

A Calm uses Provider Accounts to connect to cloud platforms (AWS, Azure, GCP, Xi) and on-premises hypervisors (AHV, VMware). Blueprints can target multiple providers simultaneously — e.g., deploying frontend VMs on AWS and backend databases on on-premises Nutanix. Calm abstracts provider differences through a unified blueprint language.

Q90 What is NCM (Nutanix Cloud Manager)?

A NCM (formerly Nutanix Cost Governance, formerly Beam) is Nutanix's cloud cost management solution. NCM provides multi-cloud cost visibility, rightsizing recommendations, reserved instance optimization, policy-driven governance, and chargeback reporting. It helps organizations reduce cloud waste and optimize spending across AWS, Azure, GCP, and Nutanix Private Cloud.

Q91 What is a Calm Project?

A A Calm Project defines the scope, resource limits (quotas), users/groups, and cloud provider accounts available for application deployments. Projects provide multi-tenancy in Calm — different teams get their own projects with separate budgets and environments. All Calm objects (blueprints, apps, runbooks) belong to a project.

Q92 How does Calm integrate with ITSM tools?

A Calm integrates with ITSM platforms (ServiceNow, Jira) via REST API webhooks and Calm's built-in REST task type. When a user requests an application from the marketplace, Calm can automatically create a ServiceNow change request, wait for approval, then proceed with deployment — enabling full ITIL-compliant change management for self-service provisioning.

Section 10: LCM, Upgrades & Troubleshooting

Q93 What is LCM (Life Cycle Manager)?

A LCM is Nutanix's automated infrastructure update management tool. It discovers available updates for AOS, AHV, firmware (BMC, BIOS, disk, NIC), NCC, Foundation, and other components. LCM performs rolling upgrades with minimal disruption, handles upgrade ordering and dependencies, and provides a single interface for all cluster component updates.

Q94 What is a rolling upgrade in Nutanix?

A A rolling upgrade updates one node at a time while the cluster remains operational. Before upgrading a node, its VMs are live-migrated to other nodes. The node is then upgraded and rebooted. VMs migrate back and the process continues to the next node. This ensures zero planned downtime for running workloads during upgrades.

Q95 What is Foundation in Nutanix?

A Foundation is Nutanix's imaging and provisioning tool used to install AOS and the hypervisor on bare-metal nodes. Foundation can be used for new cluster deployments, re-imaging existing nodes, and bulk deployments. It runs as a VM (Foundation VM) or application that discovers Nutanix hardware and automates the imaging workflow.

Q96 What is NCC and how do you run it?

A NCC (Nutanix Cluster Check) is a health check framework. Run NCC from Prism (Settings > Health > Run NCC) or via NCLI: 'ncli health-check run-all'. NCC runs checks for hardware health, configuration best practices, cluster performance, and software versions. Results show Pass/Warning/Fail with remediation recommendations.

Q97 How do you troubleshoot a node that is not coming up?

A Steps: (1) Check physical connectivity (power, network cables, IPMI). (2) Access IPMI/iDRAC for console access. (3) Check CVM logs at /home/nutanix/data/logs/. (4) Verify network connectivity: 'ping ' from another CVM. (5) Check cluster status: 'cluster status' from any CVM. (6) Review NCC health check output. (7) Contact Nutanix Support with cluster log bundle.

Q98 What command checks cluster services status on a CVM?

A Run 'cluster status' from the CVM shell to check all cluster services and their running state across all nodes. For individual services, use 'genesis status' to check genesis (service orchestrator) status. 'allssh cluster status' runs the command across all CVMs simultaneously for a cluster-wide view.

Q99 How do you collect log bundles for Nutanix Support?

A In Prism, navigate to Settings > Log Collector. Select the time range, choose relevant services (all or specific), and click Collect. The log bundle is created as a tar.gz file downloadable from Prism. Alternatively, use NCLI: 'ncli cluster collect-logs'. For urgent issues, use 'logbay' tool on the CVM for advanced log collection.

Q100 What is the Nutanix Support Portal?

A The Nutanix Support Portal (support.nutanix.com) provides: case management (open/track support tickets), KB articles, product documentation, software downloads (AOS, AHV, NCC, tools), NCC check explanations, compatibility matrices, and the Nutanix Next Community forum. Support cases can be opened online or via phone for P1 critical issues.

Q101 What are common causes of high cluster latency?

A Common causes: (1) Overloaded SSD tier (SSD full, heavy write workload). (2) Metadata disk I/O saturation. (3) Network congestion between CVMs. (4) CVM CPU starvation (insufficient vCPU for CVM). (5) Noisy neighbor VMs. (6) Replication traffic consuming bandwidth. (7) Disk failures causing rebuild I/O overhead. Use Prism Analysis to identify the root cause.

Q102 What is the difference between AOS upgrade and AHV upgrade?

A AOS upgrade updates the Nutanix storage and management stack (CVM software) and can be done independently of the hypervisor. AHV upgrade updates the host kernel and AHV components, requiring node reboots (with VM live migration). LCM manages both; AOS upgrade typically runs before AHV upgrade. AOS and AHV have separate version compatibility matrices.

Section 11: Nutanix Cloud Platform & Xi/NC2

Q103 What is Nutanix Cloud Clusters (NC2)?

A NC2 (Nutanix Cloud Clusters, formerly Xi Frame and then Nutanix Clusters on AWS/Azure) allows running Nutanix AOS and AHV natively on bare-metal cloud instances (AWS i3en.metal, Azure bare-metal). NC2 provides a consistent Nutanix experience in public clouds, enabling cloud bursting, DR, and hybrid cloud workload mobility.

Q104 What cloud platforms does NC2 support?

A NC2 currently supports AWS (using i3en.metal bare-metal instances) and Microsoft Azure (using Azure bare-metal instances). NC2 nodes are managed through Prism Central alongside on-premises clusters, providing a unified hybrid cloud management experience with consistent policies, networking, and data protection.

Q105 What is Nutanix Frame?

A Nutanix Frame (formerly Cloudpaging) is a cloud-hosted virtual desktop and application delivery platform. Frame delivers Windows and Linux desktops/apps from AWS, Azure, or GCP to any browser. It is a DaaS (Desktop as a Service) solution that complements VDI workloads running on Nutanix on-premises infrastructure.

Q106 What is Nutanix Era?

A Nutanix Era is a Database as a Service (DBaaS) platform for managing databases (Oracle, SQL Server, PostgreSQL, MySQL, MongoDB, MariaDB) running on Nutanix. Era automates database provisioning (1-click), patching, backup, cloning (time-machine clones), and lifecycle operations, reducing database management overhead significantly.

Q107 What is the Nutanix Time Machine in Era?

A The Era Time Machine provides continuous data protection for databases, capturing log-based incremental changes every 30 minutes (default) plus daily snapshots. It enables point-in-time recovery to any second within the retention period. Clones can be created from any Time Machine snapshot for dev/test or DR purposes in minutes.

Q108 What is Nutanix Karbon?

A Nutanix Karbon (now NKE – Nutanix Kubernetes Engine) is a managed Kubernetes service that simplifies deploying, managing, and scaling Kubernetes clusters on Nutanix. NKE handles cluster provisioning, node OS lifecycle, Kubernetes version upgrades, persistent volume management via CSI, and integrates with Calm for GitOps-based app deployment.

Q109 What is NKE (Nutanix Kubernetes Engine)?

A NKE is Nutanix's enterprise Kubernetes platform. It provides: one-click Kubernetes cluster deployment, automated node lifecycle management, built-in monitoring (Prometheus/Grafana), persistent storage via Nutanix CSI driver, private registry support, Helm chart integration, and kubectl access. NKE clusters are managed through Prism Central.

Q110 What is Nutanix CSI Driver?

A The Nutanix Container Storage Interface (CSI) driver provides Kubernetes pods with persistent storage backed by Nutanix Volumes (block) or Nutanix Files (NFS). It supports dynamic provisioning, volume snapshots, volume cloning, and storage class customization. The CSI driver is automatically installed and managed by NKE or manually deployed on any Kubernetes cluster.

Q111 What is the Nutanix Cloud Platform (NCP) vision?

A The Nutanix Cloud Platform envisions a fully software-defined hybrid multicloud infrastructure where compute, storage, networking, security, and cloud services are consumed uniformly from a single control plane (Prism Central) regardless of whether workloads run on-premises, at colocation, or in public clouds – delivering true cloud freedom.

Q112 What is Nutanix Unified Storage?

A Nutanix Unified Storage consolidates Block (Volumes/ABS), File (Files/FSVM), and Object (Objects/S3) storage services under a single Nutanix platform. It eliminates separate NAS, SAN, and object storage silos, reduces hardware costs, simplifies management, and provides consistent data services and protection across all storage types.

Section 12: Advanced Topics & Best Practices

Q113 What is Nutanix AHV Turbo?

A AHV Turbo is a next-generation storage I/O path for AHV that uses a ring-buffer mechanism to bypass traditional kernel I/O overhead. It significantly reduces I/O latency for VMs on all-flash nodes by reducing the number of software layers in the I/O path, delivering performance closer to native NVMe speeds for latency-sensitive workloads.

Q114 What is vNUMA in Nutanix AHV?

A Virtual NUMA (vNUMA) exposes NUMA topology to VMs, allowing the guest OS to optimize memory access across NUMA nodes. AHV supports vNUMA for large VMs (>8 vCPUs) on multi-socket hosts. Enabling vNUMA can reduce memory access latency for memory-bandwidth-intensive workloads like large databases and in-memory analytics.

Q115 What are the Nutanix sizing best practices for VDI?

A For VDI on Nutanix: use all-flash nodes for persistent desktops, enable inline deduplication and compression, use linked clones for non-persistent desktops, allocate 2–4 vCPUs and 2–4 GB RAM per desktop session, leverage Nutanix Files for user profile storage, use Prism Central for multi-cluster VDI capacity planning, and test with Login VSI.

Q116 What is the Sizer tool in Nutanix?

A Nutanix Sizer is an online sizing and configuration tool (sizer.nutanix.com) used by Nutanix sales engineers and partners to design cluster configurations for specific workloads (VDI, OLTP databases, ROBO, general virtualization). It calculates node count, model selection, storage efficiency, and generates BOM (Bill of Materials) for proposals.

Q117 What is the Nutanix Compatibility Matrix?

A The Nutanix Compatibility Matrix (available at portal.nutanix.com) lists supported combinations of AOS versions, AHV versions, hypervisor versions (ESXi, Hyper-V), hardware models, firmware versions, and 3rd party software integrations. Always check the compatibility matrix before planning upgrades or adding new components to ensure supported configurations.

Q118 What are NUMA best practices in Nutanix?

A Best practices: (1) Match host NUMA topology – keep VMs smaller than one NUMA domain where possible. (2) Enable vNUMA for VMs larger than one NUMA node. (3) Use Memory Overcommit carefully – excess overcommit causes cross-NUMA memory access. (4) For database VMs, allocate vCPUs in multiples matching host NUMA node core count.

Q119 What is the recommended CVM memory size?

A CVM memory should not be reduced below minimum supported values (typically 32 GB for all-flash nodes). Adding more RAM to the CVM improves metadata caching and deduplication performance. For large clusters with many VMs and heavy deduplication workloads, 64 GB CVM RAM is recommended. Consult the Nutanix CVM Memory Configuration guide for specifics.

Q120 What is the Pulse feature in Nutanix?

A Pulse is Nutanix's telemetry and proactive support feature. When enabled, it sends anonymized cluster health data (hardware events, alerts, performance metrics, NCC results) to Nutanix Support. Nutanix uses Pulse data for proactive support outreach (alerting customers to issues before they escalate), improving products, and generating support insights.

Q121 What is Nutanix Move?

A Nutanix Move is a workload mobility tool that migrates VMs from VMware vSphere, Hyper-V, AWS, and Azure to Nutanix AHV with minimal downtime. Move performs a block-level copy of VM disks to AHV while the source VM remains running, then does a final cutover sync with a brief outage. It supports bulk migrations with scheduling and validation reports.

Q122 What are the key differences between NCP-MCI and NCP-MCA certifications?

A NCP-MCI (Multicloud Infrastructure) focuses on core Nutanix infrastructure skills: AOS, AHV, Prism, storage, networking, data protection, and cluster management — the exam for infrastructure/system administrators. NCP-MCA (Multicloud Automation) focuses on Calm, NCM, API automation, X-Play, and cloud governance — targeting automation engineers and cloud architects.

Section 13: Nutanix APIs & Integration

Q123 What APIs does Nutanix provide?

A Nutanix provides: Prism Element REST API v1/v2/v3 (cluster-level), Prism Central REST API v3 (multi-cluster), AHV REST API (hypervisor management), Nutanix Volumes API (iSCSI management), Era API (database services), and Calm API (application automation). All APIs use HTTPS with JSON payloads and basic authentication or API keys.

Q124 What is the Nutanix v3 API?

A The Nutanix v3 API (intentful API) uses an intent-based model — you declare the desired state of objects (VMs, storage, networks) rather than issuing imperative commands. v3 is the primary API for Prism Central operations and supports VMs, categories, images, security policies, projects, and complex query/list operations with filters.

Q125 How do you authenticate with the Nutanix REST API?

A Nutanix REST APIs use HTTP Basic Authentication (Base64-encoded username:password) in the Authorization header. For production automation, use dedicated service accounts with minimal RBAC permissions. Prism Central also supports API keys for token-based authentication. All API calls must use HTTPS (port 9440) to protect credentials in transit.

Q126 What tools can you use to interact with Nutanix APIs?

A Common tools: Postman (GUI API testing), curl (command-line), Python with requests library, Ansible nutanix.ncp collection, Terraform Nutanix provider, PowerShell (Invoke-RestMethod), and Nutanix's own Python SDK (ntnx_api). Nutanix also provides API Explorer in Prism for interactive API testing without external tools.

Q127 What is the Nutanix Terraform Provider?

A The Nutanix Terraform Provider (registry.terraform.io/providers/nutanix/nutanix) enables Infrastructure as Code management of Nutanix resources using HashiCorp Terraform. It supports creating/managing VMs, storage containers, networks, clusters, categories, images, and protection policies via declarative HCL configuration files with state management.

Q128 What is the Nutanix Ansible Collection?

A The nutanix.ncp Ansible Collection provides Ansible modules for automating Nutanix infrastructure: VM creation, network management, storage configuration, cluster operations, and Calm blueprint deployments. It uses the Prism Central v3 API and supports idempotent operations — safe to run repeatedly for configuration enforcement.

Q129 What is the Prism API Explorer?

A The API Explorer is a built-in Swagger UI in Prism Central (accessible at https://:9440/api/nutanix/v3/api_explorer) that documents all available v3 API endpoints, request/response schemas, and allows interactive testing directly from the browser. It is invaluable for learning API structure and testing calls before implementing automation.

Q130 How do you create a VM using the Nutanix v3 API?

A Send a POST to <https://:9440/api/nutanix/v3/vms> with a JSON body specifying: spec.name, spec.resources (num_vcpus, memory_size_mib, disk_list, nic_list), spec.cluster_reference, and metadata (kind: vm). The response includes the task UUID to track provisioning status via GET [/tasks/{uuid}](#). The VM appears in Prism once the task completes.

Q131 What is the Nutanix SDK?

A Nutanix provides Python SDKs (`ntnx_vmm_py_client`, `ntnx_prism_py_client`, etc.) via PyPI that wrap REST API calls with typed objects, retry logic, and authentication handling. SDKs simplify automation development by providing language-native interfaces instead of raw HTTP calls and include sample code for common operations.

Q132 What are webhooks used for in Nutanix?

A Webhooks in Nutanix (configured in Prism Pro Alerts) send HTTP POST notifications to external systems (Slack, PagerDuty, ServiceNow) when alerts are triggered. X-Play playbooks can also use REST tasks as webhooks to trigger external automation workflows. Webhooks enable Nutanix to integrate with existing monitoring and ITSM ecosystems.

Section 14: VDI & End User Computing on Nutanix

Q133 What VDI platforms does Nutanix support?

A Nutanix supports leading VDI platforms including Citrix Virtual Apps and Desktops (CVAD), VMware Horizon, Microsoft Azure Virtual Desktop (AVD), and Nutanix Frame (cloud-hosted DaaS). Nutanix is a preferred infrastructure platform for VDI due to its deduplication, flash-optimized I/O, and linear scalability for desktop workloads.

Q134 What is Nutanix Frame and how is it different from traditional VDI?

A Nutanix Frame is a cloud-hosted DaaS solution that delivers Windows apps/desktops from cloud providers (AWS, Azure, GCP) to any browser. Unlike traditional VDI running on-premises infrastructure, Frame is consumed as a service with per-user pricing, elastic scaling, and no infrastructure to manage — users access desktops via a browser URL.

Q135 What storage optimizations are critical for VDI on Nutanix?

A Key VDI optimizations: Enable inline deduplication (identical OS blocks are deduplicated across all VMs — typically 40–70% space savings). Use compression (15–30% additional savings). Deploy linked clones for non-persistent desktops (multiple VMs sharing a base disk image). Use all-flash nodes for boot storm performance and consistent desktop experience.

Q136 What is a boot storm and how does Nutanix handle it?

A A boot storm occurs when many VMs start simultaneously (morning logon peak), generating massive random I/O. Nutanix handles boot storms using: all-SSD read cache (serving boot reads from local SSD), deduplication (identical OS pages served from single cached block), and the SSD tier absorbing peak IOPS that HDD-based storage cannot handle.

Q137 How do you size Nutanix for Citrix VDI?

A Use Nutanix Sizer tool with workload type = Citrix XenDesktop. Input: number of users, user profile (task worker/knowledge worker/power user), concurrent usage %, desktop OS (Windows 10/11), Gold image size, and storage efficiency settings. Sizer calculates node count, CPU:user ratio, memory per node, and estimated storage savings.

Q138 What is Nutanix AFS/Files integration with VDI?

A Nutanix Files provides SMB share storage for VDI user profiles (Citrix User Profile Management, FSLogix for AVD/Horizon). Using Nutanix Files for profiles keeps profile storage on the same Nutanix cluster as the desktops, eliminating external NAS dependencies, providing high performance with close data locality, and simplifying management.

Q139 What is a persistent vs. non-persistent VDI desktop?

A Persistent desktops are dedicated VMs that retain user changes between sessions (applications, settings saved permanently) — higher storage cost. Non-persistent (pooled) desktops are reset to a golden image after each session — much lower storage cost (50–80% savings with deduplication). Nutanix supports both; non-persistent benefits most from deduplication.

Q140 What AHV feature benefits VDI workloads during maintenance?

A AHV Live Migration allows moving running desktop VMs to other nodes during host maintenance (upgrades, hardware replacement) without disconnecting active user sessions. This ensures users experience no interruption during infrastructure maintenance windows, which is critical for 24/7 VDI environments.

Q141 What monitoring is available for VDI on Nutanix?

A Prism Pro provides VM-level performance dashboards showing CPU/memory/IOPS per desktop VM. Nutanix integrates with Citrix Director and VMware vROps for end-to-end VDI monitoring. Third-party tools like Login VSI measure VDI performance baselines. Prism Pro anomaly detection can alert on unusual desktop performance patterns.

Q142 What network requirements are important for VDI on Nutanix?

A Key network requirements: 10 GbE minimum (25 GbE preferred) for storage backplane between nodes. Dedicated management VLAN for CVMs. VLAN separation for desktop traffic. For WAN delivery, Citrix SD-WAN or VMware SD-WAN can optimize ICA/PCoIP/Blast Extreme protocol delivery. Low-latency network is critical for interactive desktop performance.

Section 15: Exam Preparation & Key Concepts Summary

Q143 What are the core topics covered in the NCP-MCI 6.x exam?

A NCP-MCI 6.x covers: Plan and Deploy Infrastructure (cluster setup, node configuration), Configure and Manage (storage, networking, VMs, security), Implement Data Protection (snapshots, replication, DR), Monitor and Troubleshoot (NCC, Prism analytics, log collection), and Plan for Scalability (capacity planning, upgrades, LCM). The exam has approximately 75 questions with a 120-minute time limit.

Q144 What are the prerequisites for the NCP-MCI certification?

A Prerequisites: Complete the Nutanix Hybrid Cloud Fundamentals (HCF) training course and pass the Nutanix Certified Associate (NCA) exam (or have an active NCA certification). Practical hands-on experience with Nutanix clusters is strongly recommended. Nutanix provides a free Test Drive environment and HPOC (Hosted POC) labs for practice.

Q145 What study resources are available for NCP-MCI?

A Official resources: Nutanix University NCP-MCI Training course (hands-on labs), Nutanix Bible (nutanixbible.com), Nutanix documentation portal (portal.nutanix.com), NCC check documentation, Nutanix Next Community forums, and practice exams. Third-party: Udemy courses, YouTube channels (Nutanix TV), and study groups on LinkedIn/Reddit (r/Nutanix).

Q146 What is the Nutanix Bible?

A The Nutanix Bible (nutanixbible.com) is a comprehensive technical reference authored by Steven Poitras, a Nutanix co-founder. It provides deep technical explanations of Nutanix architecture, DSF internals, data path, data structures, I/O handling, and upgrade procedures. It is considered essential reading for NCP-MCI exam preparation.

Q147 How long is the NCP-MCI certification valid?

A NCP-MCI certification is valid for 2 years from the date of passing. To recertify, pass the current version of the NCP-MCI exam or complete Nutanix-approved continuing education. Nutanix University provides recertification paths that may include updated courses as Nutanix releases new AOS versions and features.

Q148 What is the NCA exam and how does it relate to NCP-MCI?

A The Nutanix Certified Associate (NCA) is an entry-level certification covering fundamental Nutanix concepts: HCI basics, Prism overview, storage architecture, networking, and data protection fundamentals. NCA is a prerequisite for NCP-MCI. Passing NCA demonstrates foundational knowledge before tackling the more in-depth NCP-MCI infrastructure exam.

Q149 What are the most commonly tested NCP-MCI scenarios?

A High-frequency exam topics: Configuring storage containers and enabling data efficiency features, setting up Protection Domains and DR replication, troubleshooting cluster health using NCC, performing rolling upgrades with LCM, configuring AHV networks and VLANs, understanding RF2 vs RF3 trade-offs, and using Prism analytics for performance investigation.

Q150 What is the significance of RF2 vs RF3 in exam context?

A RF2: 2 data copies, tolerates 1 node/disk failure, requires minimum 3 nodes, ~50% usable storage. RF3: 3 data copies, tolerates 2 simultaneous failures, requires minimum 5 nodes, ~33% usable storage. Exam questions test when to use each (RF2 for general use, RF3 for critical databases requiring higher fault tolerance) and capacity calculation implications.

Q151 How do you calculate usable storage with Nutanix efficiency features?

A Formula: Raw Capacity $\times$ (1/RF) $\times$ Compression Ratio $\times$ Dedup Ratio $\times$ Erasure Coding savings = Usable Capacity. Example: 100 TB raw, RF2 (50%), 2:1 compression, 1.5:1 dedup = $100 \times 0.5 \times 2 \times 1.5 = 150$ TB effective. Use Nutanix Sizer for accurate calculations. Exam questions test understanding of how each feature impacts usable capacity.

Q152 What are the top 10 things to remember for NCP-MCI exam day?

A 1) RF2=3 nodes min, RF3=5 nodes min. 2) CVM handles all storage I/O. 3) LCM for all upgrades. 4) NCC for health checks. 5) Protection Domains for snapshots/DR. 6) Prism Central manages multiple clusters. 7) Flow for micro-segmentation on AHV. 8) Calm for application automation. 9) RBAC roles in Prism Central. 10) Data locality = VM data on local node SSD. Review these concepts thoroughly before the exam.

General

Q153 What does 'scale-out' mean in Nutanix context?

A Scale-out means adding more nodes to an existing cluster to increase capacity and performance linearly. Unlike scale-up (adding more CPU/RAM to existing servers), scale-out distributes workload across more nodes. Nutanix supports non-disruptive scale-out — adding a node through the Expand Cluster wizard in Prism without any downtime or manual data migration.

Q154 What is the Nutanix Distributed Metadata Store?

A The Distributed Metadata Store uses a Paxos-based consensus algorithm to store cluster metadata (block maps, extent group locations, VM configurations) across all CVMs. This eliminates single-point-of-failure metadata controllers (like traditional storage array controllers), providing fault tolerance and consistent metadata even during node failures.

Q155 What is an extent in Nutanix DSF?

A An extent is the basic addressable unit of storage in the DSF, typically 1 MB in size. VMs write data in extents, which are distributed across nodes for RF compliance. Extent groups (collections of extents on a disk) are the physical storage containers. The metadata store tracks which extent groups hold which VM data extents.

Q156 What is Nutanix Acropolis?

A Acropolis is the software layer within AOS responsible for VM and storage management on AHV clusters. It includes the Acropolis Master (cluster-wide VM orchestration), Acropolis Slave (per-node VM management), and the Distributed Storage Fabric. The acropolis service runs inside the CVM and manages VM lifecycle operations.

Q157 What is the difference between Nutanix CE and commercial AOS?

A Nutanix Community Edition (CE) is a free version of AOS for non-production use, labs, and testing. It runs on commodity hardware but lacks commercial support, SLA guarantees, and some enterprise features. Commercial AOS includes support contracts, all enterprise features (Calm, Leap, NCM), hardware-specific optimizations, and software updates via LCM.

Q158 What is a Nutanix HBA (Host Bus Adapter) configuration?

A Nutanix nodes use direct-attached storage (DAS) — local SSDs and HDDs connected via SATA, SAS, or NVMe controllers on the host motherboard. There is no external HBA or SAN fabric. The CVM accesses local disks via pass-through from the hypervisor. This DAS model eliminates SAN complexity and provides predictable, low-latency local storage performance.

Q159 What is AHV UEFI support?

A AHV supports UEFI (Unified Extensible Firmware Interface) boot for VMs, enabling modern OS features including Secure Boot, larger disk support (>2 TB boot disks), and compatibility with modern Windows and Linux distributions. UEFI is configured per-VM during creation in Prism. Legacy BIOS is also supported for older guest OSes.

Q160 What is the Nutanix cluster VIP (Virtual IP)?

A The Cluster Virtual IP is a floating IP address that always points to the Prism Element leader CVM. If the CVM running Prism leader fails or is upgraded, the VIP automatically moves to another CVM. This ensures uninterrupted access to the Prism web interface and API endpoint, even during rolling upgrades or CVM failures.

Q161 What is Nutanix iSCSI Data Services IP?

A The iSCSI Data Services IP is a cluster-level virtual IP used by Volume Groups for iSCSI target access. External hosts connecting to Nutanix iSCSI LUNs connect via this IP rather than individual CVM IPs. The Data Services IP ensures that if a CVM fails, iSCSI connections can reconnect transparently through the remaining CVMs.

Q162 What is the process for replacing a failed disk in Nutanix?

A When a disk fails, Prism shows an alert and marks the disk as bad. The failed disk is removed (hot-swap capable in most platforms). A new disk of the same type/size is inserted — DSF automatically detects the new disk, adds it to the storage pool, and begins rebuilding (reconstructing) the missing data replicas from remaining copies across the cluster.

Q163 What is the impact of a CVM being offline?

A If a CVM is offline on one node, that node's VMs continue running under the hypervisor, but their I/O is redirected to CVMs on other nodes via the network (remote I/O path). Performance may degrade slightly due to network I/O overhead vs. local I/O. The DSF automatically routes I/O to available CVMs, maintaining cluster availability.

Q164 What is Nutanix Flow Security Central?

A Flow Security Central is a Nutanix security posture management product that provides visibility into security compliance, threat detection, and policy recommendations across Nutanix and cloud environments. It analyzes network flows, identifies anomalies, and integrates with SIEM tools to help security teams identify and respond to potential threats.

Q165 What is the Nutanix Intelligent Operations feature?

A Intelligent Operations (part of Prism Pro) uses machine learning to analyze historical performance data and predict future capacity exhaustion, performance bottlenecks, and efficiency optimization opportunities. It provides What-If modeling, automated anomaly detection, and JIT performance recommendations to proactively manage cluster resources.

Q166 What are the key metrics to monitor in a Nutanix cluster?

A Critical metrics: CPU utilization (cluster and per-host), memory utilization (cluster, host, VM), storage capacity (raw, usable, used), IOPS (read/write), I/O latency (target <1ms for all-flash), throughput (MB/s), CVM CPU/memory usage, disk health, network bandwidth utilization, and NCC health check score. Monitor via Prism dashboards and set alert policies.

Q167 What is a Nutanix Alert Policy?

A Alert Policies in Prism define thresholds for metrics that trigger alerts (Warning or Critical). Policies can send email notifications, SNMP traps, or syslog messages. Custom alert policies can be created for VM-specific or cluster-wide metrics. Alert policies integrate with X-Play playbooks for automated remediation when thresholds are exceeded.

Q168 What is the Nutanix Foundation Central?

A Foundation Central is a centralized deployment tool for managing large-scale Nutanix cluster deployments and re-imaging operations. It extends Foundation capabilities to remote sites, allowing IT teams to image multiple clusters in different locations simultaneously from a central interface, without requiring on-site technicians for software deployment.

Q169 What is IPMI and how is it used in Nutanix?

A IPMI (Intelligent Platform Management Interface) is the out-of-band management interface (also called BMC – Baseboard Management Controller) on each Nutanix node. IPMI provides power management, hardware health monitoring, remote console access, and PXE boot capabilities independent of the OS. IPMI is used for node management when the OS/hypervisor is unresponsive.

Q170 What is the Nutanix Prism API rate limit?

A Nutanix Prism REST APIs have rate limiting to prevent overload: typically ~100 requests/second per client IP. Automation scripts should implement exponential backoff and retry logic for 429 (Too Many Requests) responses. Batch API operations (using list endpoints rather than individual GET calls) reduce API call volume for bulk inventory or monitoring queries.

Q171 What is the difference between warm and cold migration in Nutanix?

A Warm (live) migration moves a running VM to another host with minimal downtime (<1 second cutover) using memory pre-copy. Cold migration moves a powered-off VM to another host – no memory state is transferred, just disk location metadata updates (since storage stays in DSF). Cold migration is instant in Nutanix because disk data doesn't move.

Q172 What is Nutanix AHV PC (PC-managed AHV)?

A When AHV clusters are registered with Prism Central, VM management operations (create, migrate, snapshot, power ops) can be performed from PC rather than individual PE instances. Prism Central provides centralized VM inventory, cross-cluster workload visibility, policy-based placement (VM categories), and unified search across all managed AHV clusters.

Q173 What are Nutanix placement policies?

A VM Placement Policies in Prism Central enforce rules about which clusters, hosts, or host groups VMs can run on. Policies can be affinity (VMs must run on specific hosts) or anti-affinity (VMs must not share a host). Policies are applied via VM categories and enforced by the AHV scheduler during VM placement and live migration decisions.

Q174 What is Nutanix's approach to shadow clones?

A Shadow clones are read-only cached copies of VM disk data automatically created by DSF when multiple VMs on different nodes frequently read the same data blocks (e.g., a shared base image). Shadow clones reduce cross-node read traffic by serving reads locally, significantly improving performance for VDI environments with many VMs sharing a common OS image.

Q175 What is the Nutanix Global Deduplication?

A Global deduplication (post-process) deduplicates data across the entire container, not just per-node. It uses SHA-1 fingerprinting to identify identical 4KB/16KB chunks across all VMs in a container. The fingerprint table is stored in Cassandra. Best suited for VDI deployments where many identical OS image blocks exist across hundreds of VMs.

Q176 What is the difference between container-level and global deduplication?

A Container-level deduplication (AOS 4.x era) operated within a single container's data set. Global deduplication (introduced later) operates across the entire storage tier on a node, finding duplicates between multiple containers on the same node. Global dedup provides higher savings rates for environments with mixed workloads sharing common data patterns.

Q177 What is AOS's approach to write I/O handling?

A All writes from VMs go to the local CVM, which writes data to the local SSD (OpLog – write buffer). The OpLog acknowledges the write to the VM immediately for low latency. Data is then drained from OpLog to the Extent Store asynchronously. Simultaneously, the CVM replicates the OpLog entry to N-1 other CVMs (for RF compliance) before acknowledging.

Q178 What is the OpLog in Nutanix?

A The OpLog is a write buffer stored on SSD within the CVM's local storage. All incoming VM writes are staged in the OpLog first for fast acknowledgment (sub-millisecond latency). The OpLog acts as a write-combining buffer that coalesces random small writes into larger sequential writes before draining to the Extent Store, improving overall write performance.

Q179 What happens when the OpLog is full?

A If the OpLog fills (due to heavy write workloads overwhelming drain speed), new writes are throttled and go directly to the Extent Store, increasing write latency. Nutanix alerts on high OpLog utilization. Resolution: check CVM CPU/disk health, reduce write workload, or upgrade to all-flash (faster drain). NCC alerts on OpLog_full conditions.

Q180 What is AES in Nutanix context?

A AES (Acropolis Elastic Scheduling) is AHV's dynamic CPU scheduling mechanism. Unlike traditional hypervisor credit-based schedulers, AES uses a work-conserving, latency-optimized scheduler that reduces vCPU scheduling latency for latency-sensitive workloads (databases, real-time apps). AES is automatically used for all VMs on AHV without configuration.

Q181 What is memory ballooning in AHV?

A Memory ballooning (via the virtio-balloon driver in the guest) allows AHV to reclaim unused memory from VMs that are not actively using all allocated RAM. The balloon driver inflates (allocates memory within the guest, forcing the guest to release less-used pages to the host) or deflates (releases memory back to the VM) based on host memory pressure.

Q182 What is transparent huge pages (THP) in AHV?

A Transparent Huge Pages (THP) is a Linux kernel feature used by the AHV host to map memory in 2MB pages instead of 4KB pages, reducing TLB misses and improving memory access performance for large VMs. AHV enables THP by default for the host OS. For certain latency-sensitive applications (databases), disabling THP inside the guest VM may improve performance.

Q183 What is the significance of 10 GbE vs 25 GbE in Nutanix deployments?

A 10 GbE provides 10 Gbps bandwidth per port (~1.2 GB/s), sufficient for most hybrid workloads. 25 GbE (2.5x bandwidth) is recommended for all-flash clusters where NVMe drives can saturate 10 GbE storage backplane during heavy I/O or DSF rebuild operations. High-density VDI and database clusters benefit significantly from 25 GbE network upgrades.

Q184 What RAID level equivalent does RF2 represent?

A RF2 is conceptually similar to RAID-1 (mirroring) but distributed across multiple nodes rather than two disks. Each data block has exactly 2 copies on different nodes (and preferably different physical blocks). Unlike RAID-1 which is limited to 2 disks, RF2 across N nodes provides N-1 resilience for individual disk failures across different nodes.

Q185 What is a storage pool in Nutanix?

A A Storage Pool is the aggregation of all physical disk capacity (SSD + HDD) across all nodes in the cluster into a single shared storage resource. In older AOS versions, Storage Pools were explicitly configured; in modern AOS, the pool is automatically managed by DSF as a single default pool. All containers draw capacity from this shared pool.

Q186 What is the Nutanix platform model NX vs. DX/XC/DX?

A NX series: Nutanix-branded hardware (manufactured by SuperMicro) sold and supported entirely by Nutanix. Dell XC series: Dell-branded Nutanix-compatible hardware with Dell support for hardware and Nutanix support for AOS software. HPE DX series: HPE-branded Nutanix hardware. Lenovo ThinSystem HX: Lenovo-branded. All run identical AOS software.

Q187 What is Nutanix's approach to data integrity?

A DSF uses end-to-end checksums (CRC32 or SHA-1) stored in metadata for every extent group. During every read, the checksum is verified against stored data. Detected corruption is automatically corrected using the RF replica or EC parity data. Periodic scrubbing (via Curator) proactively detects and repairs silent data corruption on all stored data.

Q188 What is the Nutanix cluster expansion process?

A Process: (1) Physically rack the new node and connect network/power. (2) Image the node using Foundation (if bare metal). (3) In Prism Element: Settings > Expand Cluster > Add Node. (4) Prism discovers unconfigured nodes. (5) Configure IP addresses, hypervisor, and options. (6) Confirm and execute. (7) DSF automatically rebalances data to include new node capacity.

Q189 What is Nutanix Xi?

A Xi (pronounced 'Zi') is Nutanix's family of cloud services — Xi Cloud Services — including Xi Leap (cloud DR), Xi Frame (cloud VDI), and Xi IoT. The Xi brand represents Nutanix's extension from on-premises HCI to hybrid cloud services, running Nutanix software in Nutanix-operated cloud data centers for cloud-bursting and DR use cases.

Q190 What is the Nutanix SPDK (Storage Performance Development Kit) integration?

A Nutanix uses SPDK concepts in its I/O path for NVMe devices, bypassing kernel I/O overhead using user-space drivers and polling-mode drivers. SPDK integration reduces CVM CPU cycles per I/O operation and achieves lower latency for NVMe-based all-flash nodes. This is part of the AHV Turbo initiative to maximize NVMe performance.

Q191 What is the maximum cluster size for Nutanix?

A Standard Nutanix clusters support up to several hundred nodes (the specific maximum varies by AOS version and is documented in release notes — typically 256+ nodes per cluster for current AOS versions). For larger environments, Nutanix recommends multiple clusters managed by Prism Central rather than single mega-clusters for operational manageability.

Q192 What is AHV Live Migration bandwidth behavior?

A AHV live migration uses available bandwidth between source and destination nodes over the storage network. For VMs with heavy memory write rates (dirty pages), more migration iterations may be needed. AHV throttles migration bandwidth to avoid impacting production workloads. The migration is completed when dirty page rate drops below the network copy rate (convergence).

Q193 What is the CVM startup sequence?

A CVM startup sequence: (1) Hypervisor boots (AHV kernel + OVS). (2) CVM VM starts (KVM guest). (3) Genesis starts inside CVM (service orchestrator). (4) Genesis starts all AOS services in order: Zookeeper, Cassandra, Stargate, Acropolis, Cerebro, Prism, etc. (5) When all services are healthy, the CVM joins cluster quorum and becomes operational.

Q194 What is the Nutanix pre-upgrade check process?

A Before any upgrade (AOS/AHV/firmware via LCM), Nutanix automatically runs pre-upgrade NCC health checks to verify the cluster is healthy. Checks include: no existing alerts, sufficient free space, all nodes reachable, services healthy, compatible upgrade path. Upgrades are blocked if critical pre-checks fail, preventing upgrades on already-degraded clusters.

Q195 What is the significance of the metadata ring in Nutanix?

A The metadata ring is the Cassandra distributed hash ring that partitions metadata storage across all CVMs. Each CVM owns a portion of the metadata keyspace (a token range). With RF3 metadata replication, each metadata entry is stored on 3 consecutive nodes in the ring. The ring provides linear metadata scalability as nodes are added.

Q196 What is a Nutanix category key vs. category value?

A In Nutanix, a Category is a key-value pair. The Key is the dimension (e.g., 'Environment', 'AppType', 'Tier'). The Value is the specific attribute (e.g., 'Production', 'Database', 'Web'). Categories are assigned to VMs, images, or subnets. Prism Central uses categories to apply policies (Flow security, PC policies, Calm governance) to grouped resources.

Q197 What is Flow Network Visualization?

A Flow Network Visualization in Prism Central displays actual network traffic flows between VM groups (categories) as a visual traffic map. It shows which VMs are communicating with which other VMs, highlighting unexpected connections. This discovery mode helps administrators understand current communication patterns before implementing Flow security policies to avoid inadvertently blocking legitimate traffic.

Q198 What is Nutanix Prism Central scale-out?

A Prism Central can be deployed as a single VM (small/large) or as a 3-VM scale-out cluster for high availability and increased capacity. The scale-out PC cluster uses Elasticsearch and other distributed services across 3 PC VMs, providing HA for the management plane. Scale-out PC is recommended for environments with >3 clusters or heavy Prism Pro analytics usage.

Q199 What is the Nutanix AHV CPU overcommit ratio recommendation?

A Nutanix recommends a CPU overcommit ratio of up to 4:1 (vCPU:pCPU) for general workloads. For VDI, ratios up to 6:1 are common with task workers. For databases and latency-sensitive applications, maintain 1:1 or 2:1 overcommit. Excessive overcommit increases vCPU ready time (scheduling wait), degrading application performance. Monitor vCPU ready % in Prism.

Q200 What is memory overcommit in Nutanix AHV?

A AHV supports memory overcommit – allocating more total vRAM across VMs than physical RAM available on the host. AHV uses ballooning and swapping to manage overcommit. Best practice: limit memory overcommit to 1.0-1.25x for production workloads to avoid excessive swapping. For VDI with memory optimization, moderate overcommit (1.5-2x) can be acceptable.

Q201 What is the Nutanix Support SLA structure?

A Nutanix Support offers multiple SLA tiers: Basic (8x5 support, next business day hardware replacement), Professional (24x7 support, 4-hour hardware replacement), Ultimate (24x7, 2-hour hardware, dedicated team). For P1 (cluster down) issues, all tiers provide 24x7 phone support. SLA response times vary by priority (P1: 30-60 min, P2: 2-4 hours, P3/P4: next business day).

Q202 What is the Nutanix ROBO (Remote Office/Branch Office) solution?

A Nutanix ROBO deployments address small sites with 1-2 node clusters (with witness VM for quorum). Nutanix offers 1-node and 2-node configurations with a remote witness VM (hosted on cloud or central datacenter) to maintain quorum without requiring 3 nodes locally. ROBO nodes are managed centrally via Prism Central, reducing on-site IT requirements.

Q203 What is a Witness VM in Nutanix?

A A Witness VM is a lightweight VM deployed on a remote site (or cloud) that maintains quorum for a 2-node ROBO cluster. Without 3 nodes, a 2-node cluster cannot self-determine which node should be authoritative during a split-brain scenario. The Witness VM breaks the tie by maintaining quorum state, allowing the cluster to continue operating when one node is unavailable.

Q204 What is the Nutanix hardware qualification process?

A Nutanix qualifies hardware platforms through rigorous testing of CPU models, memory configurations, NIC types, storage controllers, and drive models. Qualified hardware is listed in the Hardware Compatibility List (HCL) at portal.nutanix.com. Using non-HCL hardware or components voids support. LCM firmware upgrades are only provided for HCL-listed components.

Q205 What is the Nutanix cluster health score?

A The cluster health score in Prism is a percentage (0-100%) representing overall cluster health based on NCC check results. Green (>85%) = healthy, Yellow (60-85%) = warning, Red (<60%) = critical issues. The health score aggregates hardware, software, configuration, and performance check results. Navigate to Prism Health tab for detailed per-check status.

Q206 What is the Nutanix move tool migration process?

A Nutanix Move migration steps: (1) Deploy Move VM on target Nutanix AHV cluster. (2) Configure source provider (vCenter, Hyper-V, AWS, Azure) and target Nutanix cluster. (3) Create migration plan: select VMs, map networks and datastores. (4) Start migration (disk sync begins). (5) Test mode option. (6) Execute cutover – source VM pauses, final sync completes, VM powers on at Nutanix target.

Q207 What is PC (Prism Central) VM sizing?

A Prism Central sizing: Small (1 VM, 4 vCPU, 16 GB RAM, 500 GB disk) – up to 2,500 VMs. Large (1 VM, 10 vCPU, 30 GB RAM, 2.5 TB disk) – up to 25,000 VMs. Scale-out (3 VMs, each 10 vCPU, 30 GB RAM) – up to 150,000 VMs. Prism Pro analytics require Large or scale-out PC. PC sizing depends on number of clusters and VMs managed.

Q208 What is the Nutanix 'dark site' LCM approach?

A Dark site LCM enables Nutanix clusters without internet access to use LCM for upgrades. An administrator downloads the LCM bundle (from portal.nutanix.com on an internet-connected machine), hosts it on an internal HTTP web server, and configures LCM to use this local URL instead of the Nutanix CDN. All LCM operations then work without outbound internet connectivity.

Q209 What is Nutanix's multi-cluster VM inventory in Prism Central?

A Prism Central provides a unified VM inventory across all registered clusters. The Virtual Machines view shows all VMs from all clusters in a single table with filtering, sorting, and bulk operations. Administrators can perform power operations, migrations, snapshot operations, and VM configuration changes from PC without navigating to individual cluster Prism Elements.

Q210 What is the difference between a container's reserved and advertised capacity?

A Reserved capacity in a container sets aside a guaranteed minimum storage allocation, preventing other containers from using it. Advertised capacity (overcommit) allows thin-provisioned VMs to see more capacity than physically available, since most VMs don't use their full allocated disk size. Nutanix monitors actual usage and alerts when real consumption approaches physical limits.

Q211 What are the key differences between Nutanix AOS versions?

A Major AOS version milestones: 5.x introduced PC scale-out, Flow, Leap DR. 6.0 introduced AHV Turbo, enhanced Kubernetes (NKE v2), PC Dark Site LCM. 6.5+ introduced improved NC2 support, Files v4, Objects v3, and enhanced security features. Each version's release notes (on portal.nutanix.com) list new features, fixed issues, and upgrade paths.

Q212 What is the Nutanix Cloud Governance capability?

A Cloud Governance (part of NCM) provides policy-based controls over cloud resource consumption: budget alerts, mandatory tagging policies, approval workflows for new cloud deployments, resource expiration policies, and idle resource detection. It prevents uncontrolled cloud sprawl by enforcing organizational policies on who can deploy what resources in which cloud accounts.

Q213 What is the difference between Nutanix backup and replication?

A Backup creates point-in-time copies of VM data stored locally (Protection Domain snapshots) or exported to backup software (Veeam, Commvault). Replication sends snapshot copies to a remote Nutanix cluster for DR purposes. Backup focuses on data recovery from corruption/deletion. Replication focuses on site-level DR. Both are often used together for comprehensive data protection.

Q214 What is NGT self-service restore?

A With NGT installed and Nutanix Files configured, end users can restore their own deleted/modified files from NFS snapshots via a Windows Explorer extension (right-click > Restore Previous Versions) without IT involvement. This reduces helpdesk tickets for accidental file deletions and empowers users to perform immediate self-service recovery from File Server snapshots.

Q215 What is Nutanix Era multi-cluster support?

A Era can manage databases across multiple Nutanix clusters registered to the same Era Server. A single Era instance provides a unified view of all databases across clusters, enables cross-cluster Time Machine cloning, and manages DB patching centrally. Era Server is deployed as a VM on one cluster and can register additional clusters as compute targets for DB deployments.

Q216 What is the Nutanix compatibility between AOS and AHV versions?

A AOS and AHV have a strict compatibility matrix — not all AOS versions are compatible with all AHV versions. Before upgrading, check the Nutanix Upgrade Path tool on portal.nutanix.com to identify the correct upgrade sequence. Typically, AHV must be upgraded to the compatible version before or during the AOS upgrade, following LCM's recommended upgrade order.

Q217 What is the Nutanix Insights feature?

A Nutanix Insights is a cloud-based analytics service that aggregates telemetry data from multiple customer clusters (via Pulse) to provide industry benchmark comparisons, best-practice recommendations, and proactive health advisories. It helps Nutanix Support proactively identify patterns across the installed base to predict and prevent widespread issues.

Q218 What is iSCSI multipathing in Nutanix?

A Nutanix Volume Groups support iSCSI multipathing — external hosts can connect to a Volume Group via multiple iSCSI paths (multiple CVM IPs) for both redundancy and load balancing. If one CVM is unavailable, the iSCSI initiator automatically fails over to another CVM path. Configure multipathing via host iSCSI initiator settings pointing to multiple Nutanix CVM IPs.

Q219 What is Nutanix AHV's support for GPU passthrough?

A AHV supports GPU passthrough (PCIe passthrough) for VMs requiring direct GPU access (AI/ML workloads, GPU-accelerated VDI with NVIDIA vGPU, rendering). Requires: IOMMU enabled in BIOS, GPU listed in AHV compatibility, and PCIe passthrough configured in AHV. For NVIDIA vGPU (shared GPU), the vGPU Manager (NVIDIA grid license required) runs inside AHV.

Q220 What are the phases of Nutanix cluster startup?

A Cluster startup phases: (1) Physical hosts power on, hypervisor boots. (2) CVMs start (as VM guests on each host). (3) Genesis starts on each CVM, discovers peers. (4) Zookeeper quorum establishes across CVMs. (5) Cassandra ring initializes, syncs metadata. (6) Stargate starts I/O services. (7) Acropolis Master elected, VM management available. (8) Prism webserver starts, GUI accessible.

Q221 What is the Nutanix Smart DR capability?

A Smart DR (part of Nutanix Leap) uses policy-based automation to define protection SLAs — instead of manually configuring individual Protection Domains, administrators define protection policies (RPO target, retention) and assign them to VM categories. Any VM tagged with the category is automatically protected according to the policy, simplifying large-scale DR management.

Q222 What is the Nutanix AHV backup integration approach?

A AHV-based VMs are backed up by third-party backup solutions (Veeam, Commvault, HYCU, Rubrik, Cohesity) via the Nutanix REST API and/or XTRACT agents. Backup vendors use Nutanix's Changed Block Tracking (CBT) API for incremental backups — only changed disk blocks since the last backup are transferred, dramatically reducing backup windows and bandwidth.

Q223 What is Changed Block Tracking (CBT) in Nutanix?

A CBT is an API capability in Nutanix that tracks which disk blocks have changed since a specific snapshot. Backup software calls the CBT API to retrieve only the blocks that changed since the last backup job, enabling efficient incremental backup without full disk scans. CBT dramatically reduces backup time and data transfer for large VMs with small daily change rates.

Q224 What is the purpose of the Nutanix Health Summary dashboard?

A The Health Summary dashboard in Prism provides a consolidated view of cluster health across: hardware (node/disk status), storage (capacity, efficiency), network, data protection, and VM health. It shows active Critical/Warning alerts, NCC check results, cluster health score, and recent events — serving as the first stop for daily cluster health reviews.

Q225 What is the Nutanix Prism REST API authentication using API keys?

A Prism Central supports service account API keys for authentication in API integrations. Navigate to PC Settings > API Keys to generate a long-lived API key token. This token is passed in the 'x-ntnx-api-key' HTTP header instead of Basic Auth credentials, providing more secure automation integration without embedding plaintext passwords in scripts.

Q226 What is the NCP-MCI exam passing score?

A The NCP-MCI passing score is typically 3000 out of 6000 points (approximately 50%). The exam uses a scaled scoring model where questions have different point values based on difficulty. Nutanix does not publicly disclose the exact passing threshold, but exam preparation materials suggest aiming for 70%+ accuracy on practice questions to ensure a comfortable passing margin.

Q227 What is the NCP-MCI exam format?

A The NCP-MCI exam consists of approximately 75 multiple-choice and multi-select questions delivered via Pearson VUE (online proctored or test center). Time limit is 120 minutes (~1.6 minutes per question). Questions cover scenario-based troubleshooting, configuration decisions, and conceptual knowledge. The exam is available in English and Japanese.

Q228 What is the Nutanix training path for NCP-MCI?

A Recommended path: (1) Nutanix Hybrid Cloud Fundamentals (free, online — core concepts). (2) Nutanix HCI Administration (instructor-led or self-paced — hands-on labs). (3) NCA exam (associate-level prerequisite). (4) Nutanix NCP-MCI training course (instructor-led). (5) Practice exams and lab time (HPOC or Test Drive). (6) Schedule and pass NCP-MCI exam.

Q229 What is the Nutanix Test Drive?

A Nutanix Test Drive is a free, browser-based demo environment that provides limited-time access to a live Nutanix cluster with guided scenarios for exploring Prism, VM management, storage operations, and data protection features. It requires no installation or hardware and is available at nutanix.com/test-drive — useful for NCP-MCI candidates to explore the Prism interface hands-on.

Q230 What is HPOC (Hosted Proof of Concept)?

A HPOC is Nutanix's reserved lab environment available to Nutanix partners and customers for POC evaluations and training. HPOC provides dedicated Nutanix cluster access in Nutanix data centers for a specified duration. NCP-MCI training courses often include HPOC lab time for hands-on exercises including cluster configuration, VM management, and troubleshooting scenarios.

Q231 What is the difference between a single-node cluster and multi-node cluster in Nutanix?

A A single-node cluster (1-node CE or edge deployment) has no data redundancy — there is no second node to replicate data to. It is suitable only for non-production, testing, or specific edge use cases. Multi-node clusters (3+ nodes for RF2) provide data redundancy, HA, and linear scalability. Production deployments always require a minimum of 3 nodes.

Q232 What is the purpose of the Nutanix Genesis service?

A Genesis is the service orchestrator and watchdog process running on each CVM. It starts and monitors all other AOS services (Zookeeper, Cassandra, Stargate, Acropolis, Cerebro, Prism, etc.), restarts failed services, manages cluster join/leave operations, handles CVM upgrades, and provides the service health framework that ensures cluster services remain operational.

Q233 What is the Nutanix Prism Central REST API pagination?

A Nutanix v3 API uses offset-based pagination for list operations. Each list request returns a 'metadata' block with 'total_matches', 'offset', and 'length'. To paginate, increment 'offset' by 'length' in subsequent requests until all records are retrieved. For large environments (10,000+ VMs), implement pagination in automation scripts to avoid API timeouts from single large responses.

Q234 What is the Nutanix Acropolis File Services (AFS) healing process?

A When a Nutanix Files server VM (FSVM) fails, the remaining FSVMs automatically take over its NFS/SMB share responsibilities using a distributed share ownership model. Client connections briefly reconnect to surviving FSVMs. The failed FSVM is automatically restarted by AHV HA on a healthy node, then rejoins the file server cluster — typically restoring full capacity within minutes.

Q235 What are the Nutanix Files deployment components?

A Nutanix Files deploys as a cluster of 3+ FSVMs (File Server VMs) running on AHV. Each FSVM provides NFS/SMB services and is backed by dedicated Nutanix storage containers for data and metadata. An external Load Balancer VIP handles client connections. FSVMs communicate with CVMs for storage I/O and use Active Directory for SMB authentication and Kerberos support.

Q236 What is the Nutanix Objects minimum deployment?

A Nutanix Objects requires a minimum of 3 nodes in the cluster for deployment. It deploys as a set of Object Store worker pods running as VMs/containers on the Nutanix cluster. Objects uses the underlying DSF for storage, inheriting RF2/RF3 data protection. The Objects management interface is accessible via Prism Central. Minimum cluster capacity recommendation is 1 TB raw per worker node.

Q237 What is S3 compatibility in Nutanix Objects?

A Nutanix Objects provides Amazon S3-compatible API endpoints, allowing any application that supports S3 (AWS SDK, s3cmd, Veeam, Commvault, Veritas, etc.) to use Nutanix Objects as a storage target without application modification. Standard S3 operations — PUT, GET, DELETE, LIST, multipart upload, bucket policies, versioning, lifecycle rules — are fully supported.

Q238 What is Nutanix Karbon Registry?

A NKE includes a built-in private Docker container registry (Harbor-based) for storing container images locally within the Nutanix environment. This is valuable for air-gapped environments that cannot access public container registries (Docker Hub, Quay.io). The private registry supports image vulnerability scanning, RBAC, and is integrated with NKE cluster deployment for local image pulls.

Q239 What are the vGPU modes supported on AHV?

A AHV supports NVIDIA vGPU in these modes: vComputeServer (vCS – for AI/ML compute workloads, full GPU partitioning), vPC (Virtual PC – for high-end knowledge worker VDI), vApp (for server-based virtualized apps), vWS (Quadro virtual workstation – for CAD/engineering graphics VDI). Each mode has different frame buffer allocation and use case optimization characteristics.

Q240 What is the Nutanix AHV SR-IOV support?

A SR-IOV (Single Root I/O Virtualization) allows a single physical NIC to be partitioned into multiple Virtual Functions (VFs), each assignable to a VM for near-native network performance. AHV supports SR-IOV for high-throughput networking workloads (NFV, HPC, low-latency trading). SR-IOV bypasses the OVS software switch, so Flow micro-segmentation policies do not apply to SR-IOV VFs.

Q241 What is the Nutanix AHV nested virtualization support?

A AHV supports nested virtualization (running a hypervisor inside a VM) on hosts where the physical CPU supports Intel VT-x with EPT or AMD-V with RVI. Nested virtualization is useful for developer environments, testing hypervisor configurations, and running vSphere simulations. Enable per-VM with the 'Enable CPU passthrough' option in Prism (exposes host CPU features to guest).

Q242 What is the Nutanix Security Advisory process?

A When a security vulnerability is discovered in Nutanix software, Nutanix's Product Security team publishes a Security Advisory on the Nutanix Support Portal with: CVE ID, CVSS score, affected AOS/AHV versions, description, and mitigation steps. Critical CVEs receive emergency patches. Customers are encouraged to enable email notifications for new security advisories.

Q243 What is the Nutanix AOS update cadence?

A Nutanix follows a predictable release cadence: major LTS (Long Term Support) releases every 18-24 months with 18-month support lifecycle, and STS (Short Term Support) releases more frequently with 6-month support. LTS releases are recommended for production environments. Patch releases (x.y.z) address critical bugs and security issues within each major version.

Q244 What are the hardware requirements for running Prism Central?

A Prism Central VM requirements: Small deployment — 4 vCPU, 16 GB RAM, 500 GB storage. Large deployment — 10 vCPU, 30 GB RAM, 2.5 TB storage. PC must run on a Nutanix AHV or ESXi cluster (it cannot run on Hyper-V). The hosting cluster must be running a compatible AOS version. PC should run on a separate cluster from the clusters it manages in production environments.

Q245 What is Nutanix's support for VMware vSphere integration?

A When running vSphere (ESXi) on Nutanix, the DSF presents storage as NFS datastores to ESXi hosts. All VMware features (vMotion, HA, DRS, vSphere Replication) work natively. Nutanix provides a vCenter Plugin for Prism integration within vCenter. Management can be done from both vCenter and Prism Element simultaneously. Note: Nutanix Calm, Flow, and AHV-specific features require AHV.

Q246 What is the VMware vCenter plugin for Nutanix?

A The Nutanix vCenter Plugin integrates Prism Element management capabilities directly into the vCenter Server web interface. Administrators can view Nutanix storage health, manage containers, check alerts, and access Prism directly from within vSphere Web Client without switching to separate browser tabs. The plugin is installed from Prism Settings > vCenter Registration.

Q247 What is the Nutanix support for Microsoft Hyper-V?

A Nutanix supports Microsoft Hyper-V as a hypervisor, presenting DSF storage as SMB 3.0 file shares to Hyper-V hosts. Standard Hyper-V features (Live Migration, Cluster Shared Volumes, Hyper-V Replica) work with Nutanix storage. Management is via Prism Element alongside SCVMM/Hyper-V Manager. Note: AHV-specific features (Flow, Leap DR for AHV) are not available on Hyper-V.

Q248 What is a Nutanix protection policy in Prism Central?

A Protection Policies in Prism Central (Leap) define replication schedules and retention rules as reusable templates. A policy specifies: local snapshot schedule, remote replication schedule, RPO target, and retention count. Policies are assigned to VM categories — all VMs in that category are automatically protected by the assigned policy, replacing per-VM Protection Domain configuration.

Q249 What is the role of zookeeper in Nutanix?

A Zookeeper is used within Nutanix AOS for distributed coordination, leader election (for services like Acropolis Master and Prism Leader), and configuration management. It maintains a quorum across CVMs and ensures that critical cluster services have a single leader. Losing Zookeeper quorum (>50% CVMs offline) causes cluster management operations to stop.

Q250 What is the Cassandra service in Nutanix?

A Cassandra is used within Nutanix as the distributed metadata store database. It stores extent map metadata (where VM data blocks are located), fingerprint (deduplication) tables, and other cluster metadata. Cassandra runs on each CVM and uses RF3 internally for its own metadata to ensure metadata is never lost even with two CVM failures.

Q251 What is the Stargate service?

A Stargate is the Nutanix data I/O manager service running on each CVM. It handles all VM storage I/O operations: reading from and writing to DSF, enforcing data locality, managing the SSD/HDD tiering, handling replication for RF, and managing erasure coding operations. Stargate is the most performance-critical service in the CVM.

Q252 What is the Curator service in Nutanix?

A Curator is the Nutanix cluster management and maintenance service. It runs periodically to: rebalance data across nodes (after node addition/removal), perform disk scrubbing, clean up stale snapshots, optimize storage layout, run deduplication and compression scans, and ensure RF compliance after failures. Curator runs as a background process to maintain cluster health.

Q253 What is the Cerebro service?

A Cerebro is the Nutanix replication and data protection service. It manages Protection Domain operations: scheduling and executing snapshots, replicating snapshot data to remote sites, managing DR failover/failback workflows, and maintaining consistency group operations. Cerebro coordinates with remote site Cerebro instances for async/synchronous replication.

Q254 What is the Acropolis Master?

A The Acropolis Master is the elected leader among CVM Acropolis services that makes cluster-wide VM scheduling decisions: VM placement on hosts, live migration orchestration, HA restart coordination, and resource reservation enforcement. The master election uses Zookeeper for consensus. If the master CVM fails, a new master is elected automatically.

Q255 What is a managed vs. unmanaged VLAN in Nutanix AHV?

A Managed VLAN: Nutanix manages DHCP and IPAM for VMs in this network — administrators define a subnet, IP pool, gateway, and DNS. VMs get IPs automatically from Nutanix's built-in DHCP server. Unmanaged VLAN: Nutanix just tags VM traffic with the VLAN ID — all DHCP/IP management is handled by external infrastructure. Unmanaged VLANs are used when integrating with existing IPAM systems.

Q256 What is the purpose of the extent store in Nutanix?

A The Extent Store is the persistent storage layer in DSF where actual VM data (extents) are stored on physical disks. The Extent Store manages disk I/O, handles SSD/HDD tier placement, executes data efficiency operations (compression, erasure coding), and maintains data integrity via checksums on all stored extents for silent data corruption detection.

Q257 What is the default CVM vCPU and memory allocation?

A Default CVM allocation varies by node model: typically 4–8 vCPUs and 16–32 GB RAM for hybrid nodes. All-flash nodes often receive 12–20 vCPUs and 32–64 GB RAM to handle higher I/O throughput and metadata operations. CVM resources are reserved and not available to user VMs. Changing CVM resources requires guidance from Nutanix Support.

Q258 What is Nutanix Karbon's container networking?

A NKE (Nutanix Kubernetes Engine) uses Flannel as the default CNI (Container Network Interface) plugin for pod-to-pod networking. It creates an overlay network (VXLAN) across all Kubernetes worker nodes. For production deployments, Calico or other CNI plugins can be deployed post-cluster creation for network policy enforcement and BGP routing.

Q259 What is the process for decommissioning a Nutanix node?

A To decommission a node: (1) Ensure cluster has sufficient capacity after removal. (2) Evacuate VMs via live migration (or let HA handle it). (3) Use 'Remove Host' in Prism (Hardware > Hosts > Remove Host). (4) Nutanix automatically rebalances data off the node. (5) Wait for data rebalance completion. (6) Physically remove the node from the rack.

Q260 What is the Nutanix cluster resiliency status?

A Cluster resiliency status (visible in Prism Home) indicates the cluster's ability to tolerate additional failures. Status values: OK (full RF2/RF3 met), Critical (one failure already occurred, at-risk), Data Unavailable (RF violated, some data inaccessible). Resiliency status updates within minutes of any hardware event, guiding urgency for replacement parts.

Q261 What is the Nutanix UVM (User VM)?

A A UVM (User Virtual Machine) is any VM created and managed by a user on the Nutanix cluster, as distinguished from the CVM (which is a system VM). UVMs run application workloads (databases, web servers, desktops, etc.) on the hypervisor layer. The CVM provides storage services to UVMs but is not counted in UVM licensing calculations.

Q262 What is the Nutanix AHV oVirt/RHV compatibility?

A AHV is based on KVM but uses libvirt and a custom management stack (Acropolis) rather than oVirt/RHEV. AHV is not compatible with oVirt management tools directly. AHV is managed exclusively through Prism/aCLI/REST API. However, standard KVM tools (virsh) can be used on AHV hosts for troubleshooting, though direct VM management via virsh is not supported in production.

Q263 What is the difference between Nutanix community and paid support?

A Community support (free) includes Nutanix Next forum, documentation portal, KB articles — suitable for CE users and self-service troubleshooting. Paid support (Basic/Professional/Ultimate) provides direct access to Nutanix Support engineers via phone/portal, hardware replacement SLAs, proactive monitoring, and patch downloads. Production deployments require paid support contracts.

Q264 What is the Nutanix AOS rolling upgrade verification?

A After each node upgrade in a rolling upgrade, Nutanix verifies: CVM services are healthy, all NCC pre-checks pass, data is accessible, RF compliance is maintained, and no active alerts exist before proceeding to the next node. If verification fails, the upgrade pauses and alerts fire — the administrator must resolve issues before the upgrade can continue to the next node.

Q265 How does Nutanix handle a split-brain scenario?

A Split-brain occurs when cluster nodes cannot communicate (network partition). Nutanix uses Zookeeper quorum — the partition with >50% of nodes maintains cluster operations. The minority partition stops I/O operations to prevent data corruption. In 2-node clusters, the Witness VM serves as the quorum tiebreaker. The cluster automatically recovers when network connectivity restores.

Q266 What is the Nutanix storage summary in Prism?

A The Storage Summary in Prism shows: Total Raw Capacity, Total Used Capacity, Logical Used Capacity (before efficiency), Storage Efficiency ratio (deduplicated/compressed/EC savings), Tier breakdown (SSD/HDD usage), and per-container utilization. Hover over the capacity donut chart to see breakdown by container. Monitor this daily to predict capacity exhaustion.

Q267 What are the Nutanix networking requirements for a new cluster deployment?

A Networking requirements: 1 management network (for CVMs, hypervisor, IPMI), 1 storage/backplane network (CVM-to-CVM DSF traffic, can overlap management for small deployments), 1+ VM network (VLANs for VM traffic). Each node needs 2+ NICs (bonded). 10/25 GbE switches recommended. Dedicated VLAN for IPMI/BMC. Jumbo frames (MTU 9000) recommended for storage network.

Q268 What is the Nutanix AHV bonding mode recommendation?

A Nutanix recommends active-backup bonding (mode 1) for simplicity — no switch configuration required, provides NIC redundancy. For higher throughput, balance-slb (mode 2) provides load balancing across uplinks without LACP. For maximum performance, balance-tcp (LACP/802.3ad) requires switch LACP configuration but provides true per-flow load balancing. Active-backup is the default for new deployments.

Q269 What is the Nutanix CVM network configuration?

A Each CVM requires: 1 internal (hypervisor-to-CVM) network interface (192.168.5.0/24 default for AHV, using a bridge). 1 external network interface on the management/storage VLAN with a static IP address. The CVM IP must be reachable from other CVMs and from the management workstation. Prism Central must be able to reach all CVM IPs on port 9440.

Q270 What is Nutanix's approach to SSD wear management?

A Nutanix tracks SSD wear level (Media Wearout Indicator / Percentage Used) via SMART data, monitored by the CVM. When an SSD approaches end-of-life wear thresholds, Nutanix alerts in Prism (NCC check: `ssd_wearout_check`). Planned replacement is recommended before SSD failure. All-flash nodes with high write workloads should plan SSD replacement cycles based on drive TBW (Terabytes Written) specifications.

Q271 What is the Nutanix AHV memory reservation behavior?

A AHV does not use memory reservations the same way ESXi does. All VM memory is thin-provisioned by default — AHV allocates physical memory as the guest OS demands it. When host memory pressure increases, AHV uses ballooning to reclaim unused guest memory. Hard reservations can be set per-VM to guarantee a VM always has its full allocated memory without ballooning.

Q272 What is the Nutanix cluster IP range planning?

A For a Nutanix cluster you need IPs for: each node's IPMI/BMC (1 per node), each node's hypervisor/AHV host (1 per node), each CVM (1 per node), cluster virtual IP (1 per cluster), iSCSI Data Services IP (1 per cluster), Prism Central VM IPs (1 or 3 for scale-out). Minimum IPs for a 3-node cluster: ~15 IPs. Careful IP planning prevents conflicts during expansion.

Q273 What is the Nutanix cluster time zone configuration?

A All CVMs in a Nutanix cluster must be synchronized to NTP servers. Configure NTP in Prism Element (Settings > NTP). Consistent time across all CVMs is critical for Cassandra metadata consistency, Zookeeper quorum, replication scheduling, log correlation, and security certificate validation. Recommended: use 3+ NTP servers with at least one being an internal stratum-2 server.

Q274 What is the Nutanix default storage replication factor for containers?

A New storage containers default to RF2 (Replication Factor 2) in most deployments. RF can be changed to RF3 for containers hosting business-critical VMs. The default container created during cluster setup is called 'default-container' with RF2. Best practice: create separate containers with appropriate RF settings for different workload tiers (production vs. dev/test).

Q275 What is Nutanix capacity runway in Prism Pro?

A Capacity Runway is a Prism Pro feature that predicts, based on historical growth trends, when the cluster will reach capacity exhaustion for CPU, memory, or storage. The runway is expressed in days (e.g., 'Storage runway: 45 days'). This allows administrators to proactively order hardware before capacity constraints impact operations, rather than reactively expanding after running out.

Q276 What are Nutanix entity charts used for?

A Entity charts in Prism Analysis provide customizable performance graphs for any cluster entity (VM, host, container, disk). Users can plot multiple metrics on the same chart (e.g., IOPS vs. latency) over custom time ranges to correlate performance events. Entity charts are used for incident investigation, capacity planning, and demonstrating SLA compliance to stakeholders.

Q277 What is the Nutanix support for Microsoft SQL Server?

A Nutanix provides validated SQL Server deployment guides. SQL Server on AHV benefits from: Volume Groups (iSCSI) for data/log LUNs with <1ms latency, AES CPU scheduler for consistent vCPU access, Era for automated SQL Server lifecycle management (provisioning, cloning, patching), NGT for VSS-consistent snapshots, and Nutanix Files for shared storage scenarios.

Q278 What is the Nutanix support for SAP HANA?

A Nutanix is SAP-certified for running SAP HANA on Nutanix NX all-flash platforms. SAP HANA on Nutanix benefits from NVMe all-flash performance, persistent memory support, and single-platform management. Nutanix provides SAP HANA sizing guides, and certification covers both HANA scale-up (single large VM) and scale-out (distributed HANA nodes) configurations.

Q279 What is the Nutanix support for Oracle databases?

A Nutanix supports Oracle databases via: Volume Groups (iSCSI) for ASM disk groups, Era for Oracle Database lifecycle management (provisioning, time-machine clones, patching), Oracle RAC support via shared Volume Groups across VMs, and NX all-flash platforms meeting Oracle's performance requirements. Nutanix provides an Oracle best practices guide with specific configuration recommendations.

Q280 What is a Nutanix DR runbook?

A A DR runbook (in Leap Recovery Plans) is an ordered set of steps defining how VMs should be failed over: power-on sequence, boot delays between stages, IP address mapping (production to DR network), test network configuration, and pre/post-failover scripts. Runbooks ensure multi-tier applications (web > app > DB) restart in the correct dependency order during DR events.

Q281 What is Nutanix's cloud-native application support?

A Nutanix supports cloud-native applications through: NKE (Kubernetes), Nutanix CSI driver for persistent storage, Objects for S3-compatible storage, integration with service meshes (Istio), GitOps support via Calm + Helm, and multi-cloud consistency. Cloud-native workloads containerized in Docker/Kubernetes can run on Nutanix using the same underlying HCI infrastructure as traditional VMs.

Q282 What is Nutanix's recommended upgrade path?

A Always: (1) Check Upgrade Path tool on portal.nutanix.com for your current and target versions. (2) Run NCC health checks and resolve alerts. (3) Upgrade NCC first (latest version). (4) Upgrade Foundation (if needed). (5) Upgrade AOS (rolling, no downtime). (6) Upgrade AHV (if needed, rolling). (7) Upgrade firmware via LCM. (8) Validate health post-upgrade. Never skip major versions.

Q283 What is the Nutanix SCMA (Security Configuration Management Automation)?

A SCMA is a built-in AOS security hardening framework that enforces CIS benchmarks and STIG compliance continuously. It monitors AHV host and CVM security configurations, auto-remediates deviations, and provides compliance reports. SCMA covers: SSH settings, password policies, audit log configuration, kernel parameter hardening, and service controls — reducing manual hardening effort.

Q284 What is the Nutanix REST API versioning strategy?

A Nutanix maintains multiple REST API versions simultaneously (v1, v2, v3) for backward compatibility. v1 is legacy (PE-only, cluster management basics). v2 added enhanced storage and VM operations. v3 (intentful API) is the current standard for Prism Central operations using intent-based spec/status model. New features are developed in v3 only; v1/v2 are maintained but not enhanced.

Q285 What is the significance of the Nutanix 5-second metric collection?

A Nutanix collects performance metrics every 30 seconds (not 5 seconds for most metrics), aggregating into 1-minute, 1-hour, and daily rollups. Some critical metrics (storage I/O) are captured more frequently. The 30-second granularity provides detailed trending for performance analysis in Prism Analysis charts. Prism Pro may use more granular data for anomaly detection.

Q286 What is Nutanix's approach to multi-tenancy?

A Nutanix provides multi-tenancy via: Prism Central Projects (resource quotas and user isolation for Calm), RBAC with cluster/category scoping, network VLANs for traffic isolation, Flow security policies for east-west isolation, separate storage containers per tenant (with optional encryption), and Nutanix Objects with per-bucket access policies. True hypervisor-level isolation uses dedicated clusters per tenant.

Q287 What is the Nutanix AHV hugepage support for VMs?

A AHV guests can be configured to use 1GB hugepages (backed by host huge pages) to improve performance for memory-intensive workloads. Hugepages reduce TLB misses and improve memory bandwidth for SAP HANA, Oracle, and in-memory databases. Configure via guest OS settings (Linux: mount hugetlbfs, Windows: Large Address Aware). The AHV host must have hugepages pre-allocated.

Q288 What are Nutanix Categories best practices?

A Best practices for Categories: (1) Define a consistent taxonomy before deployment (Environment, Tier, AppType, Owner). (2) Use category values as building blocks for Flow policies, protection policies, and PC governance. (3) Assign categories at VM creation time (via Calm blueprints or PC). (4) Avoid excessive category proliferation – keep the category schema manageable. (5) Document category purpose and ownership.

Q289 What is the Nutanix AHV PCIe passthrough use case?

A PCIe passthrough in AHV assigns a physical PCIe device (GPU, FPGA, SmartNIC, NVMe drive) directly and exclusively to a single VM, bypassing hypervisor emulation for maximum device performance. Use cases: AI/ML training VMs requiring full GPU performance, FPGA-accelerated network processing, high-frequency trading VMs needing sub-microsecond NIC latency. PCIe passthrough devices cannot be shared between VMs.

Q290 What is the Nutanix Files Tiering feature?

A Nutanix Files supports intelligent data tiering via integration with cloud object storage. Cold files (not accessed for configurable days) are automatically tiered from Nutanix Files NFS/SMB shares to AWS S3, Azure Blob, or Nutanix Objects. Data is recalled on-demand when accessed. This extends effective file storage capacity without adding Nutanix nodes for cold data.

Q291 What is Nutanix AHV maintenance mode?

A Placing an AHV host in maintenance mode (in Prism or via aCLI: 'acli host.enter_maintenance_mode') live-migrates all running VMs off the host to other cluster nodes. The host then shows as 'Entering Maintenance' then 'In Maintenance'. During maintenance mode, no VMs are scheduled on that host. Use for hardware maintenance, HBA replacement, or pre-upgrade verification.

Q292 What is the Nutanix AHV default network configuration on first setup?

A During AHV cluster setup (Foundation), a default 'default-net' network bridge (br0) is created on each host, connecting to physical uplinks. The default network uses VLAN 0 (untagged). During initial cluster configuration in Prism, administrators create named networks with appropriate VLAN tags for VM traffic. The CVM uses a dedicated internal bridge (virbr0/br0) for CVM-to-hypervisor communication.

Q293 What is the Nutanix NX platform hardware overview?

A Nutanix NX platforms include: NX-1000 series (entry-level, 1-2 nodes/block), NX-3000 series (mid-range all-NVMe or hybrid, 2-4 nodes/block), NX-5000 series (performance all-flash), NX-8000 series (high-performance compute-heavy), NX-9000 series (storage-heavy, high-capacity), and G8/G9 platform generations. Each series is optimized for different workload profiles (VDI, database, general virtualization, capacity storage).

Q294 What is the Nutanix AHV virtio driver stack?

A AHV uses virtio drivers for paravirtualized I/O performance: virtio-blk or virtio-scsi for disk I/O, virtio-net for network I/O. These drivers provide near-native performance by reducing hypervisor emulation overhead. Windows VMs require VirtIO driver installation (provided in the Nutanix VirtIO installer package). Linux VMs typically have virtio drivers built into the kernel.

Q295 What is the Nutanix AHV Q35 machine type?

A AHV supports two VM machine types: i440FX (legacy BIOS, older OS compatibility) and Q35 (modern PCIe, UEFI support, IOMMU-capable for PCIe passthrough). Q35 is recommended for all new VMs as it supports UEFI Secure Boot, larger PCIe address space, PCIe passthrough, and provides better compatibility with modern OSes (Windows 11, RHEL 8+). Machine type is configured per-VM at creation time.

Q296 What is the Nutanix Files anti-virus integration?

A Nutanix Files integrates with ICAP-based anti-virus solutions (Symantec, McAfee, Trend Micro) for real-time file scanning on SMB shares. When a file is created or accessed, Files sends the file to the AV server via ICAP protocol for scanning before allowing access. This provides enterprise-grade virus protection for file shares without requiring endpoint agents on FSVM.

Q297 What is the Nutanix capacity optimization workflow?

A Capacity optimization workflow in Prism Pro: (1) View Capacity Runway to identify when each resource (CPU/RAM/storage) will be exhausted. (2) Use What-If scenarios to model adding nodes or VMs. (3) Review Rightsizing recommendations for over-provisioned VMs. (4) Implement suggested optimizations (reclaim VM resources, enable compression/dedup). (5) Re-run What-If to validate extended runway. (6) Plan hardware orders based on lead time vs. runway.